
Characterizing Optimal Mixed Policies: Where to Intervene and What to Observe

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Abstract

Intelligent agents are continuously faced with the challenge of optimizing a policy based on what they can observe (see) and which actions they can take (do) in the environment where they are deployed. Most policies can be parametrized in terms of these two dimensions, i.e., as a function of what can be seen and done given a certain situation, which we call a *mixed policy*. In this paper, we investigate several properties of the class of mixed policies and provide an efficient and effective characterization, including optimality and non-redundancy. Specifically, we introduce a graphical criterion to identify unnecessary contexts for a set of actions, leading to a natural characterization of non-redundancy of mixed policies. We then derive sufficient conditions under which one strategy can dominate the other with respect to their maximum achievable expected rewards (optimality). This characterization leads to a fundamental understanding of the space of mixed policies and a possible refinement of the agent’s strategy so that it converges to the optimum faster and more robustly. One surprising result of the causal characterization is that the agent following a more standard approach—intervening on all intervenable variables and observing all available contexts—can be restricted to a suboptimal value in some environments, regardless of the number of interactions.

1 Introduction

Agents are deployed in complex and uncertain environments where they are bombarded with high volumes of information and are expected to operate efficiently, safely, and rationally. The discipline of causal inference (CI) offers a compelling set of tools and a language that allows one to reason with the structural invariances present in complex environments [1–5]. Whenever the causal mechanisms of an underlying environment are sufficiently well-understood, the agent can design very precise interventions, bringing a certain desired state of affairs about swiftly and cleanly (e.g., personalized medical treatments, inequality-reducing tax policies). In the field of ML, bandits and reinforcement learning (RL) constitute the *de facto* framework in which agents are designed such that a certain policy is optimized and the corresponding goals can be efficiently achieved [6–8].

There is a growing literature exploring how these two frameworks (RL and CI) are related, and how this understanding can be translated into more efficient decision-making in more challenging and realistic settings. Recently, the more explicit connection between these frameworks has been made by eliciting how causal knowledge—unobserved confounders and the causal relations between actions, contexts, and rewards—can be used to improve decision-making in a variety of settings, including for both interventional [9–11] and counterfactual [12, 13] reasoning (see also [14–17] and [18–21]). Outside more traditional RL, causal inference researchers have embraced the idea of sequential decision making in terms of conditional plans or dynamic treatment regimes, while focusing on, e.g., the identifiability of causal effects from observational data [22–27].

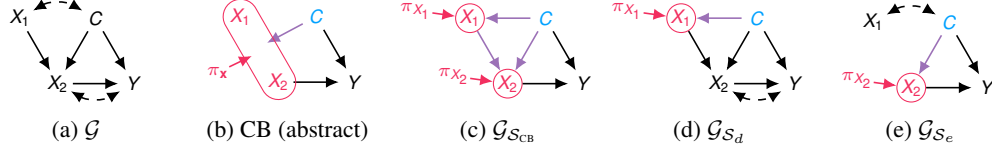


Figure 1: (a) a causal graph, (b) abstract representation of a contextual bandit policy, and (c,d,e) policy-induced graphs. Red circles for the intervened variables and, as a supplement, blue for their non-action contexts and purple for induced edges (i.e., contexts *cause* an action). π nodes are intervention indicators, which will be left implicit throughout the paper.

One of the main tasks in decision-making is to optimize the parameters associated with a specific policy. The scope of each policy is usually fixed, in the sense that the set of actions and contexts are pre-specified, *a priori*. By and large, the literature considers policies with scope that is (1) observational, where the system is allowed to evolve without any intervention; (2) fully experimental, where all the *action* variables are intervened on and all the *context* variables are observed. The former tends to be more common in CI while the latter tends to be more common in RL. A causal understanding of the world gives rise to a rich spectrum of policies with different scopes, allowing agents to choose how to interact with the environment, meaning, which variables to intervene and to observe (as a context). Against this background, we consider exploiting causal relationships for systematic decision making in the context of, so called, *mixed policies*, which consists of a set of decision rules where each rule corresponds to the way an action for an intervenable variable is determined given its contexts.

For concreteness, consider an agent deployed in an environment represented as a *causal graph* $\mathcal{G}$ (Fig. 1a), where C , $\mathbf{X} = \{X_1, X_2\}$, Y represent the context, two action variables, and the reward variable, respectively. Graphically, bidirected edges roughly represent unobserved confounders (UCs, for short) affecting both ends of the arrow. The agent’s task is to maximize the reward $\mu_\pi \doteq \mathbb{E}_\pi[Y]$ under a mixed policy (or simply, policy) $\pi \in \Pi$, where Π is a mixed policy space. A mixed policy is associated with its *scope*, called *mixed policy scope* (MPS), which specifies the variables the policy are intervening, and the variables taken into account for each intervened variables.

A standard contextual bandit (CB) optimizes a policy π_{CB} (Fig. 1b), a (stochastic) mapping from contexts to actions, which can be equally represented as a pair of decision rules $\pi_{\text{CB}} = \{\pi(x_1|c), \pi(x_2|x_1, c)\}$ (Fig. 1c). Traditionally, the policy is optimized within a restricted space Π_{CB} , characterized by policies following a scope $\mathcal{S}_{\text{CB}} = \{\langle X_1, \{C\} \rangle, \langle X_2, \{X_1, C\} \rangle\}$ that X_1 is determined by C and X_2 is decided based on C and X_1 . Unfortunately, the optimal policy $\pi_{\text{CB}}^* \doteq \arg \max_{\pi \in \Pi_{\text{CB}}} \mu_\pi$ can be suboptimal, i.e., $\mu_{\pi_{\text{CB}}^*} < \mu^*$ where μ^* is the optimal expected reward. To ground what this means, let every variable be binary and U_1 and U_2 , the UCs adjacent to X_1 and X_2 , be fair coins and ϵ be a noise over X_1 following $P(\epsilon = 1) = 0.2$. Also, let the unobserved causal mechanisms be specified as $X_1 \leftarrow U_1 \oplus \epsilon$, $C \leftarrow U_1$, $X_2 \leftarrow U_2 \oplus X_1 \oplus C$, and $Y \leftarrow (1 - (X_2 \oplus U_2)) \vee C$, where $\oplus$ is the exclusive-or operator. Since the policy determines X_2 irrelevant to U_2 and the context C is also independent to U_2 , we can elicit that $\mu_{\mathcal{S}_{\text{CB}}}^* = 0.75$. In this setting, an optimal policy intervenes only on X_1 given C , i.e., $\{\pi(x_1|c)\}$ as depicted in Fig. 1d. With $X_1 = C$, the policy suppresses the noise ϵ over X_1 and makes $X_2 = U_2$ so that its optimal expected reward in this environment is 1.0.

In the example of Fig. 1a, if $\{X_1, X_2\}$ are intervenable and $\{C, X_1\}$ can become a context, there are 15 mixed policy scopes. These different modes of interaction can be represented as induced graphs and can be classified based on two desiderata: *non-redundancy* and *optimality*. We explain these desiderata through an illustration (Fig. 2) of the four MPSes $\mathcal{S}_a = \{\}$, $\mathcal{S}_{\text{CB}}$, $\mathcal{S}_d = \{\langle X_1, \{C\} \rangle\}$, and $\mathcal{S}_e = \{\langle X_2, \{C\} \rangle\}$. We annotate their relationships with a superset symbol $\supset$, whether one scope has more actions or contexts than the other, and with a comparison symbol $\geq_\mu$ (or $=_\mu$), whether one’s optimal reward is at least as good as (or equal to) the other’s in *every* world compatible with a causal graph. The equality forms an equivalence class among scopes with respect to optimal rewards. Here *non-redundancy under optimality* (NRO; Def. 4) asks whether a scope can strictly outperform all its

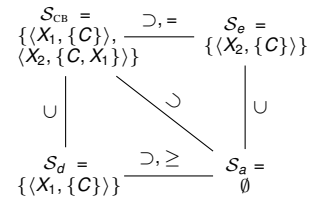


Figure 2: Relationships between the MPSes

proper subscores in one compatible world. In particular, if $\mathcal{S} \supsetneq \mathcal{S}'$ and $\mathcal{S} =_{\mu} \mathcal{S}'$, then $\mathcal{S}$ is redundant under optimality. For instance, since $\mathcal{S}_c \supset \mathcal{S}_e$ while $\mathcal{S}_c =_{\mu} \mathcal{S}_e$, the CB scope (Fig. 1c) is redundant under optimality and the CB agent wastes its resources not only for intervening on X_1 (a redundant action) but also for taking X_1 into account for X_2 (a redundant context). Furthermore, *optimality* of a scope $\mathcal{S}$ represents that there exists no other scope $\mathcal{S}'$ (not in the equivalence class of $\mathcal{S}$) such that $\mathcal{S}' \geq_{\mu} \mathcal{S}$. For example, $\mathcal{S}_d$, when optimized, is at least as good as $\mathcal{S}_a$ (i.e., $\mu_{\mathcal{S}_d}^* \geq \mu_{\mathcal{S}_a}^*$) in every environment, and can outperform it in some environments (i.e., $\mu_{\mathcal{S}_d}^* > \mu_{\mathcal{S}_a}^*$), which demonstrates that $\mathcal{S}_a$ does not meet the optimality criterion. Not every pair of scopes can be comparable: $\mathcal{S}_e$ is not comparable to $\mathcal{S}_a$ nor $\mathcal{S}_d$. After a careful examination, we can indeed show that MPSES $\mathcal{S}_c$, $\mathcal{S}_d$, $\mathcal{S}_e$ meet the optimality criterion. Here the informal optimality criterion concerns undominated equivalence classes. The definition of POMPS in Def. 5 requires one SCM in which a scope strictly outperforms all competing NRO scopes; no general equivalence between the two criteria is asserted. Both NRO and optimality in this illustration are satisfied only by $\mathcal{S}_d$ and $\mathcal{S}_e$ among all 15 scopes. This example demonstrates that an intelligent agent should judiciously intervene on a carefully chosen subset of variables with side information (context) relevant to attaining an optimal reward. More detailed account is given in Sec. A.

Contributions In this work, we investigate mixed policies with respect to their expected rewards. Our contributions are as follows. (i) We developed a graphical criterion that detects the redundancy of contexts relative to a collection of actions taking advantage of properties pertaining to optimal mixed policies. (ii) We established sufficient conditions under which one policy scope can outperform another, giving sufficient comparisons in the dominance preorder over scopes with respect to their maximum expected rewards achievable. We believe these results have practical implications for the design of intelligent agents providing the basis for efficient and effective explorations of the policy space. One fundamental implication of our analysis is that the agent following a standard approach (i.e., intervening and observing whenever possible) may be hurting itself, and, regardless of the number of interactions, will never be able to achieve an optimal performance.

Preliminaries Let us denote a variable by an uppercase letter X , whose value is denoted by its corresponding lowercase letter x . A set of variables will be denoted by a bold uppercase letter $\mathbf{X}$ with its value $\mathbf{x}$. We follow notational conventions from literature on measure theory, algebra of sets, and causal inference. We may use $\dot{\cup}$, instead of $\cup$, to emphasize the union of two disjoint sets. We use structural causal models (SCMs) [1, Ch. 7] as the semantical framework to represent an underlying environment. An SCM $\mathcal{M}$ is a quadruple $\langle \mathbf{U}, \mathbf{V}, P(\mathbf{U}), \mathbf{F} \rangle$, where $\mathbf{U}$ is a set of exogenous variables determined by factors outside the model following a joint distribution $P(\mathbf{U})$, and $\mathbf{V}$ is a set of endogenous variables whose values are determined following a collection of functions $\mathbf{F} \doteq \{f_i\}_{V_i \in \mathbf{V}}$ such that $V_i \leftarrow f_i(\mathbf{pa}_i, \mathbf{u}_i)$ where $\mathbf{pa}_i \subseteq \mathbf{V} \setminus \{V_i\}$ and $\mathbf{u}_i \subseteq \mathbf{U}$. The observational distribution $P(\mathbf{v})$ is defined as $\sum_{\mathbf{u}} \prod_{V_i \in \mathbf{V}} P(v_i | \mathbf{pa}_i, \mathbf{u}_i) P(\mathbf{u})$. Further, $do(\mathbf{X} = \mathbf{x})$ represents the operation of fixing a set $\mathbf{X}$ to a constant $\mathbf{x}$ regardless of their original mechanisms. Such intervention induces a submodel $\mathcal{M}_{\mathbf{x}}$, which is $\mathcal{M}$ with f_X replaced to x for $X \in \mathbf{X}$. Then, an interventional distribution $P_{\mathbf{x}}(\mathbf{v} \setminus \mathbf{x})$ (or also $P(\mathbf{v} \setminus \mathbf{x} | do(\mathbf{x}))$) follows from $\mathcal{M}_{\mathbf{x}}$, and is such that $P_{\mathbf{x}}(\mathbf{v} \setminus \mathbf{x}) = \sum_{\mathbf{u}} \prod_{V_i \in \mathbf{V} \setminus \mathbf{X}} P(v_i | \mathbf{pa}_i, \mathbf{u}_i) P(\mathbf{u})$.

Graphically, each SCM (model, for short) is associated with a causal diagram $\mathcal{G} = \langle \mathbf{V}, \mathbf{E} \rangle$, where each type of edge represents a different relationship among variables: (i) $X \rightarrow Y$ if X is an argument of f_Y (a direct causal relationship); and (ii) $X \leftrightarrow Y$ if for a maximal subset $\mathbf{W} \subseteq \mathbf{V} \setminus \{X\}$ such that $\mathbf{U}_{\mathbf{W}} \perp\!\!\!\perp \mathbf{U}_X$ and $\mathbf{U}_Y \not\subseteq \mathbf{U}_{\mathbf{W}}$; From the agent's perspective, only the causal graph $\mathcal{G}$ of the environment $\mathcal{M}$ is available, while its reward is validated through $\mathcal{M}$. We operate in the non-parametric setting, where no assumption about the form or shape of the pair $\langle P(\mathbf{U}), \mathbf{F} \rangle$ is made, but for the structural knowledge encoded in $\mathcal{G}$. Whenever not even $\mathcal{G}$ is known, the agent can perform active interventions to learn it; for example, see [28, 29]. We take the graph and endogenous variable domains to be finite, with nonempty domains and real-valued reward. Thus an optimal policy exists, as shown in Prop. 1. Exogenous variables may be general, with sums over them read as integrals. Graph compatibility permits structural functions to ignore listed inputs and does not impose faithfulness. Graphical separation is interpreted in an underlying latent-variable DAG compatible with the causal diagram. For graph-only existential claims, finite variable domains may be chosen as part of the constructed SCM; conclusions for prescribed cardinalities require a separate argument. In comparisons within one SCM, its domains remain fixed.

We denote by $\mathcal{G}_{\overline{\mathbf{X}}\mathbf{Z}}$ an edge subgraph of $\mathcal{G}$ which removes edges incoming to $\mathbf{X}$ and outgoing from $\mathbf{Z}$. A submodel $\mathcal{M}_{\mathbf{x}}$ can be presented as $\mathcal{G}_{\overline{\mathbf{X}}}$ with $\mathbf{X}$ fixed to $\mathbf{x}$. Hence, causal relationships among other variables are captured in $\mathcal{G} \setminus \mathbf{X}$, which is the subgraph of $\mathcal{G}$ over $\mathbf{V} \setminus \mathbf{X}$. We denote by $\mathcal{G} \langle \mathbf{V}' \rangle$ the *latent projection* of $\mathcal{G}$ onto $\mathbf{V}'$, the causal graph retaining causal relationships among $\mathbf{V}'$ [30]. We adopt familial notation, *ch*, *pa*, *an*, *de* for children, parents, ancestors, and descendants, respectively, with *Ch*, *Pa*, *An*, *De* including arguments. Our work utilizes d-separation [31, 32] and do-calculus [33], classic graphical rules to ascertain equalities between distributions (for further details, see Sec. B). Also, the omitted proofs and derivations are provided in Appendix.

2 Mixed Policies: Fundamentals & Basic Results

As discussed in the previous section, a causal understanding of the underlying world helps recognize a broad spectrum of policies with diverse scopes so as for agents to select the mode of interaction. We now formally define the space of mixed policies with the notion of mixed policy scope.

Definition 1 (Mixed Policy Scope (MPS)). Let $\mathcal{G}$ be a causal graph, Y be a specific reward variable, $\mathbf{X}^* \subseteq \mathbf{V} \setminus \{Y\}$ a set of intervenable variables, and $\mathbf{C}^* \subseteq \mathbf{V} \setminus \{Y\}$ a set of contextualizable variables. A *mixed policy scope* $\mathcal{S}$ is defined as a collection of pairs $\langle X, \mathbf{C}_X \rangle$, with at most one pair for each action X , such that (i) $X \in \mathbf{X}^*$, $\mathbf{C}_X \subseteq \mathbf{C}^* \setminus \{X\}$, and (ii) $\mathcal{G}_{\mathcal{S}}$ is acyclic, where $\mathcal{G}_{\mathcal{S}}$ is defined as $\mathcal{G}$ with edges onto X removed and directed edges from $\mathbf{C}_X$ to X added for every $\langle X, \mathbf{C}_X \rangle \in \mathcal{S}$.

For concreteness, given a causal graph $\mathcal{G}$ (Fig. 1a), the observational case is an MPS $\{\}$. An MPS $\mathcal{S}_{\text{CB}} = \{\langle X_1, \{C\} \rangle, \langle X_2, \{X_1, C\} \rangle\}$ induces a graph (Fig. 1c) while $\{\langle X_2, \{C\} \rangle\}$ induces a graph in Fig. 1e. An MPS represents a class of mixed policies that share the same graphical characteristics manifested by $\mathcal{G}_{\mathcal{S}}$, an induced graph for $\mathcal{M}_{\pi}$.

Definition 2 (Mixed Policy). Given $\langle \mathcal{G}, Y, \mathbf{X}^*, \mathbf{C}^* \rangle$ and an SCM $\mathcal{M} \sim \mathcal{G}$ with $\mathfrak{X}_Y \subseteq \mathbb{R}$, a *mixed policy* π is a realization of a mixed policy scope $\mathcal{S}$ compatible with the tuple $\pi \doteq \{\pi_{X|\mathbf{C}_X}\}_{\langle X, \mathbf{C}_X \rangle \in \mathcal{S}}$, where $\pi_{X|\mathbf{C}_X} : \mathfrak{X}_X \times \mathfrak{X}_{\mathbf{C}_X} \mapsto [0, 1]$ is a proper probability mapping.

If we consider the MPS $\mathcal{S}_{\text{CB}}$ discussed above, its mixed policy π is $\{\pi_{X_1|\{C\}}, \pi_{X_2|\{C, X_1\}}\}$, which is a specific instantiation of the parameters with respect to the corresponding scope. For readability, we may write $\{\pi(x_1|c), \pi(x_2|x_1, c)\}$. Given an underlying SCM $\mathcal{M}$, a mixed policy π induces a variant of SCM $\mathcal{M}_{\pi}$ where the function for $X \in \mathbf{X}(\pi)$ is replaced by the corresponding $\pi_{X|\mathbf{C}_X}$ (see [34] for a detailed account). Each stochastic decision rule uses a private randomizer independent of the SCM exogenous variables and of the other decision randomizers. We denote by P_{π} the joint distribution over the variables from the system under the policy π . Throughout the paper, $\mathcal{G}$, Y , $\mathbf{C}^*$, and $\mathbf{X}^*$ are oftentimes implicit including an underlying SCM $\mathcal{M} \sim \mathcal{G}$ and, thus, Π , as well.

Expected Reward We define the expected reward of a mixed policy. To begin with, we define intervened variables $\mathbf{X}(\mathcal{S}) \doteq \{X \mid \langle X, \mathbf{C}_X \rangle \in \mathcal{S}\}$ and active contexts $\mathbf{C}(\mathcal{S}) \doteq \bigcup_{X \in \mathbf{X}(\mathcal{S})} \mathbf{C}_X$. Similarly, given $\pi \sim \mathcal{S}$ (a mixed policy following the MPS), $\mathbf{X}(\pi) \doteq \mathbf{X}(\mathcal{S})$ and $\mathbf{C}(\pi) \doteq \mathbf{C}(\mathcal{S})$. Let $\mathbf{C}^- = \mathbf{C}(\pi) \setminus \mathbf{X}(\pi)$ be the *non-action* contexts. Then, the expected reward for π can be expressed as, with $\mathbf{x}$ simply denoting the value of $\mathbf{X}(\mathcal{S})$,

$$\mu_{\pi} = \sum_{y, \mathbf{x}, \mathbf{c}^-} y P_{\mathbf{x}}(y, \mathbf{c}^-) \prod_{X \in \mathbf{X}(\pi)} \pi(x|\mathbf{c}_x). \quad (1)$$

The expression separates the atomic interventional probability (first factor), which is inherent to the underlying world and not affected by the policy π , from the likelihood of a specific intervention given contexts (second factor), which is optimizable and defined by π . The expected reward can also be written focusing only on a subset of intervened variables. Given $\mathbf{X}' \subseteq \mathbf{X}(\pi)$, let $\mathbf{C}' = (\bigcup_{X \in \mathbf{X}'} \mathbf{C}_X) \setminus \mathbf{X}'$. Define $K_{\mathbf{X}'}$ directly as the distribution obtained by fixing $\mathbf{X}' = \mathbf{x}'$ and retaining every rule of π outside $\mathbf{X}'$. Its graph is a subgraph of the acyclic $\mathcal{G}_{\mathcal{S}}$, so it is well-defined. Then

$$\mu_{\pi} = \sum_{y, \mathbf{x}', \mathbf{c}'} y K_{\mathbf{x}'}(y, \mathbf{c}') \prod_{X \in \mathbf{X}'} \pi(x|\mathbf{c}_x).$$

We also write $Q'_{\mathbf{x}'} = K_{\mathbf{x}'}$ in the illustrative derivations. This notation does not require an intermediate distribution $P_{\pi \setminus \mathbf{X}'}$: restoring only some natural action mechanisms can create a directed cycle. For example, if the natural graph is $A \rightarrow B$ and the scope is $\{\langle A, \{B\} \rangle, \langle B, \emptyset \rangle\}$, removing only the intervention on B restores $A \rightarrow B \rightarrow A$.

This expression, which hides the details of uninteresting actions and contexts, is the building block to characterize mixed policies.

Optimality and deterministic mixed policy A mixed policy π is said to be *optimal* in the given environment if and only if $\mu_\pi = \mu^* \doteq \max_{\pi' \in \Pi} \mu_{\pi'}$. Restricting our attention to $\Pi_S \doteq \{\pi \in \Pi \mid \pi \sim S\}$, we define $\mu_S^* \doteq \max_{\pi' \in \Pi_S} \mu_{\pi'}$, an optimal policy π *with respect to* S . We call a mixed policy *deterministic* if, for every $\pi_{X|C_X} \in \pi$, X is determined by a function of C_X .

Proposition 1. *Given a mixed policy scope with finite endogenous domains, there exists a deterministic mixed policy, which is optimal with respect to the given scope.*

Not surprisingly at this point, a stochastic policy is no better than the best deterministic policy [35–37]. Still, this result has a particular importance to the treatment provided here due to its implications to the d-separation criterion [38], which will be instrumental and discussed in depth in Sec. 3.1. Another key implication is shown next.

Proposition 2 (Separation of Actions and Contexts). *Given an MPS S , there always exists a deterministic mixed policy $\pi \in \Pi$ such that $\mathbf{X}(\pi)$ and $\mathbf{C}(\pi)$ are disjoint and $\mu_\pi = \mu_S^*$.*

A deterministic policy gives rise to the autonomy of each action allowing them to be determined only by *non-action* contexts. For concreteness, consider the example shown in Fig. 3a. A mixed policy (Fig. 3b) includes X_2 listening to X_1 , which enables systematic coordination between X_1 and X_2 . The proposition implies that X_2 can rather listen to C (which is the context of X_1) directly (Fig. 3d). Further, in Fig. 3c, X_2 utilizes both X_1 and C . However, it is sufficient to make use of only C . By noting that the policy relative to Fig. 3d can achieve optimality, while relying on lesser information than the one relative to Fig. 3c, we investigate how to capture non-redundancy within MPSes.

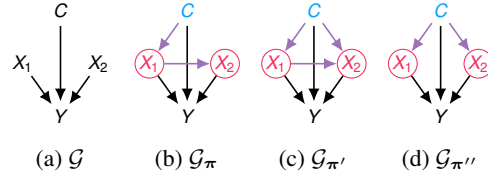


Figure 3: Given a causal graph (a), three induced graphs (b,c,d) for different mixed policies where (d) the separation is demonstrated.

3 Non-Redundant Mixed Policy

Optimizing a mixed policy involves assessments of the effectiveness of its scope so that an agent can avoid intervening or observing on unnecessary actions or contexts. Here, we define and characterize non-redundancy of MPS. We say S subsumes S' , denoted by $S' \subseteq S$, if $\mathbf{X}(S') \subseteq \mathbf{X}(S)$ and $\mathbf{C}'_X \subseteq \mathbf{C}_X$, for every $\langle X, \mathbf{C}'_X \rangle \in S'$. Further, we denote by $\pi' \subseteq \pi$, where $\pi' \sim S'$ and $\pi \sim S$ if $\pi'(x|\mathbf{c}'_x) = \sum_{\mathbf{c}''_x} \pi(x|\mathbf{c}_x) P_\pi(\mathbf{c}''_x|\mathbf{c}'_x)$, for every $X \in \mathbf{X}(S')$ where $\mathbf{C}''_X = \mathbf{C}_X \setminus \mathbf{C}'_X$. Both scopes must be valid. Strict inclusion means that the associated scope is strictly smaller. Unchanged rules are retained pointwise; for a genuinely reduced context, conditional averages on null context assignments may be completed arbitrarily.

Definition 3. Given $\langle \mathcal{G}, Y, \mathbf{X}^*, \mathbf{C}^* \rangle$, an MPS S is said to be *non-redundant* if there exists an SCM $\mathcal{M} \sim \mathcal{G}$ and $\pi \sim (S, \mathcal{M})$ such that $\mu_\pi \neq \mu_{\pi'}$ for every $\pi' \subsetneq \pi$.

The constraint on π' compares changes in actions or contexts using rules obtained by averaging the original policy under $Q = P_\pi$. For a retained action, $\pi'(x|\mathbf{c}'_x) = Q(x|\mathbf{c}'_x)$ almost surely; when its context is unchanged, its policy rule is unchanged as well. Hence, the constraint provides a basis to characterize non-redundancy of MPS utilizing well-established graphical criteria.

Theorem 1. *Consider the graph-compatible nonparametric SCM class, allowing sufficiently large finite variable domains and unused structural inputs. Let $S = \{\langle X, \mathbf{C}_X \rangle\}_{X \in \mathbf{X}}$ be an MPS and let $\mathcal{H} = \mathcal{G}_S$. S is non-redundant if and only if (i) $\mathbf{X} \subseteq \text{an}(Y)_{\mathcal{H}}$ and (ii) $(C \not\perp\!\!\!\perp Y \mid \mathbf{C}_X \setminus \{C\})$ in $\mathcal{H} \setminus \{X\}$, for every $X \in \mathbf{X}$ and $C \in \mathbf{C}_X$.*

The SCM and policy establishing non-redundancy may use vector-valued variables and independent policy randomization. The necessity direction holds for any fixed finite domains; the sufficiency statement does not assert the same result under prescribed cardinality bounds. Its proof in Sec. E constructs one SCM and one policy that distinguish all proper valid subscopes simultaneously. This is ordinary non-redundancy, not non-redundancy under optimality.

The condition (i) can be seen through rule 3 of do-calculus such that the change of the mechanism of X has a consequence on the reward Y .¹ If condition (ii) fails for a context, its rule can be averaged over that context without changing the reward distribution, as shown by the fixed-action calculation in the necessity proof. Conversely, the d-connection in (ii) supports a model where that context matters. The active path from C to Y , together with $X \leftarrow C$, explains the role of contextual information.² Consider the example in Fig. 4 where both X_1 and X_2 are ancestors of Y (condition (i)). Regarding condition (ii), C_1 being adjacent to Y , C_2 having a path to Y through X_2 , and C_3 being connected to Y as $C_3 \rightarrow C_2 \rightarrow X_1 \rightarrow Y$ demonstrate that every context is non-redundant.

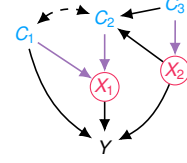


Figure 4: A non-redundant MPS

We provide the one-pass reverse-topological algorithm for obtaining the unique minimal graphical reduction of a given MPS (nr-mps, Alg. 2) in Sec. E. A graphical reduction successively deletes irrelevant contexts or nonancestor actions while preserving scope validity, as specified in Sec. E. The returned scope is non-redundant in the model class of Thm. 1, is independent of the permitted processing orders, and preserves the reward distribution under the prescribed policy reduction.

3.1 Non-Redundancy under Optimality

Non-redundancy of MPS (Def. 3) based on a stringent constraint imposed on π' is insufficient to understand, e.g., whether a context of an action would be still relevant even when $\pi \sim \mathcal{S}$ is fully-optimized. Hence, we characterize the non-redundancy of MPS under optimality, which has practical implications to an agent adapting its suboptimal policy. Recall Fig. 3c where X_2 listens to X_1 as context. We showed that the dependence is vanished under the optimality (Fig. 3d). That is, the agent would better avoid learning $\pi(x_2|c, x_1)$ at the beginning, but optimize $\pi(x_2|c)$ instead.

Definition 4 (Non-Redundancy under Optimality (NRO)). Given $\langle \mathcal{G}, Y, \mathbf{X}^*, \mathbf{C}^* \rangle$, an MPS $\mathcal{S}$ is said to be *non-redundant under optimality* if there exists an SCM $\mathcal{M}$ compatible with $\mathcal{G}$ such that $\mu_{\mathcal{S}}^* > \mu_{\mathcal{S}'}^*$ for every strictly subsumed MPS $\mathcal{S}' \subsetneq \mathcal{S}$, i.e., $\exists \mathcal{M} \sim \mathcal{G} \forall \mathcal{S}' \subsetneq \mathcal{S} (\mu_{\mathcal{S}}^* > \mu_{\mathcal{S}'}^*)$.

We will investigate a criterion more general than Thm. 1—whether, for a set of actions $\mathbf{X}' \subseteq \mathbf{X}^*$, a set of contexts $\mathbf{C}' \subsetneq \mathbf{C}_{\mathbf{X}'} \setminus \mathbf{X}'$ are relevant while taking account of deterministic relationships (Prop. 1). One approach is to characterize an opposite condition, i.e., $\mu_{\mathcal{S}}^* = \mu_{\mathcal{S}'}^*$ for $\mathcal{S}' \subsetneq \mathcal{S}$, as follows.

Proposition 3. Given an MPS $\mathcal{S}$, let $\mathbf{X}' \subseteq \mathbf{X}(\mathcal{S})$ and $\mathbf{C}' \subsetneq \mathbf{C}_{\mathbf{X}'} \setminus \mathbf{X}'$ be actions and non-action contexts of interest, respectively, and fix a mixed policy $\pi \sim \mathcal{S}$ optimal with respect to $\mathcal{S}$. For each $\mathbf{x}'$, let $K_{\mathbf{x}'}$ be the distribution obtained by retaining the rules of π outside $\mathbf{X}'$ and replacing the rules at $\mathbf{X}'$ by the constants $\mathbf{x}'$. This atomic intervention is acyclic; it does not restore the natural mechanisms at $\mathbf{X}'$. If there exist decision rules $\{\pi'(x|(\mathbf{x}' \cup \mathbf{c}') \cap \mathbf{c}_x))\}_{x \in \mathbf{X}'}$ such that

$$\mu_{\mathcal{S}}^* = \sum_{y, \mathbf{c}', \mathbf{x}'} y K_{\mathbf{x}'}(y, \mathbf{c}') \prod_{X \in \mathbf{X}'} \pi'(x|(\mathbf{x}' \cup \mathbf{c}') \cap \mathbf{c}_x), \quad (2)$$

then, $\mathbf{C}_{\mathbf{X}'} \setminus (\mathbf{C}' \cup \mathbf{X}')$ are jointly redundant to $\mathbf{X}'$ under optimality, and $\mathcal{S}' \doteq (\mathcal{S} \setminus \mathbf{X}') \cup \{\langle X, (\mathbf{X}' \cup \mathbf{C}') \cap \mathbf{C}_X \rangle\}_{X \in \mathbf{X}'}$ satisfies $\mu_{\mathcal{S}}^* = \mu_{\mathcal{S}'}^*$.

Proof. Retain π outside $\mathbf{X}'$ and use the displayed decision rules at $\mathbf{X}'$. Its scope is $\mathcal{S}'$, whose graph is a subgraph of $\mathcal{G}_{\mathcal{S}}$ because only context arrows are removed. The expected-reward formula gives its expected reward as the right-hand side of Eq. (2), hence $\mu_{\mathcal{S}'}^* \geq \mu_{\mathcal{S}}^*$. Conversely, every policy of scope $\mathcal{S}'$ can be implemented in $\mathcal{S}$ by ignoring the removed contexts, giving the reverse inequality. $\square$

To obtain a sufficient condition for Prop. 3, choose a deterministic optimal policy, whose existence follows from Prop. 1 for finite endogenous domains. Write $Q = P_{\pi}$ for this policy. We use its deterministic relationships in the d-separation criterion; an arbitrary optimal policy need not be deterministic. The graphical criterion handles deterministic mechanisms (i.e., *conditional intervention*) by excluding them appearing as common causes, e.g., $\leftarrow X \rightarrow$, in a trail [38]. This corresponds to adding those *implied* variables to the conditionals, in which we explicitly represent with an operation $[\cdot]$ for clarity. Given conditionals $\mathbf{Z}$, the implied variables with respect to $\mathbf{Z}$ is computed as follows.

¹This condition was leveraged in the atomic interventions case to establish minimality [16, 17]; see also [39].

²The relevance of contextual information has been discussed in the influence diagrams literature [40, 20]. More recently, this condition was used in the case of singleton decisions (i.e., $|\mathbf{X}(\mathcal{S})| = 1$), see [41, 42].

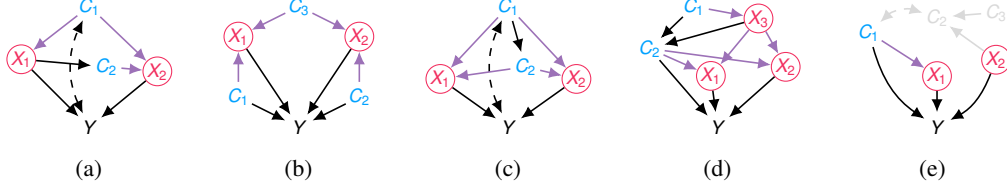


Figure 5: Causal graphs exemplifying redundancies of (a) $C_2 \rightarrow X_2$ by deterministic relationships, edges to X_1 and X_2 from (b) C_3 , (c) C_2 , (d) X_3 due to marginally or conditionally fixable contexts; (e) represents a maximal, non-redundant MPS under an optimal condition for Fig. 4.

Initially setting $\lceil \mathbf{Z} \rceil \leftarrow \mathbf{Z}$, we update $\lceil \mathbf{Z} \rceil \leftarrow \lceil \mathbf{Z} \rceil \cup \{X \in \mathbf{X}(\mathcal{S}) \mid \mathbf{C}_X \subseteq \lceil \mathbf{Z} \rceil\}$ until it is converged. For this deterministic π , every added variable is a function of $\mathbf{Z}$. Thus $\mathbf{Z}$ and $\lceil \mathbf{Z} \rceil$ generate the same information, and a d-separation statement conditional on $\lceil \mathbf{Z} \rceil$ implies the corresponding conditional independence given $\mathbf{Z}$ in P_π . For a single context $C \in \mathbf{C}_X$, put $\mathcal{H} = \mathcal{G}_\mathcal{S}$ and let $\mathbf{D}$ be the implied closure of $\mathbf{C}_X \setminus \{C\}$ using only the unchanged actions $\mathbf{X}(\mathcal{S}) \setminus \{X\}$. If $C \in \mathbf{D}$, substitute its deterministic expression directly into the rule for X . Otherwise, $(C \perp\!\!\!\perp Y \mid \mathbf{D})_{\mathcal{H} \setminus \{X\}}$ is sufficient to remove C under optimality: the fixed-action calculation averages the rule using only the retained information. The closure excludes X , which is absent from the tested graph. For instance, in Fig. 5a, C_2 as a context of X_2 is independent to Y given C_1 in a graph with X_2 removed since $\lceil \{C_1\} \rceil = \{C_1, X_1\}$ and $C_2 \leftarrow X_1 \rightarrow Y$ is not a valid trail anymore. Hence, C_2 is removable from $\mathbf{C}_{X_2}$.

Next, we illustrate contexts that unnecessarily induce correlations among actions without any implications on Y (see Sec. E.1 [43] for the derivations of the examples in Fig. 5). In Fig. 5b, both X_1 and X_2 utilize C_3 as their contexts, and $\mu_\pi = \mathbb{E}_{c_3}[\mathbb{E}_\pi[y|c_3]]$. Since there exists $c_3^* \in \arg \max_{c_3: Q(c_3) > 0} \mathbb{E}_\pi[y|c_3]$, we can derive that $\mu_\pi \leq \mathbb{E}_\pi[y|c_3^*]$. Given that c_3^* is merely a constant, new decision rules $\pi'(x_i|c_i) \doteq \pi(x_i|c_i, c_3^*)$ for $i \in \{1, 2\}$ yield the same optimal reward. A more sophisticated example is shown in Fig. 5c where a redundant context can be *fixed* conditioned on the remaining contexts. The expected reward is expressed as

$$\mu_\pi = \sum_{c_1, c_2} Q(c_2|c_1) \left(\sum_{y, \mathbf{x}} y P_\mathbf{x}(y, c_1) \pi(x_1|c_1, c_2) \pi(x_2|c_1, c_2) \right) = \sum_{c_1, c_2} Q(c_2|c_1) \mu_\pi(c_1, c_2).$$

Let c_2^* be a function taking c_1 such that $c_2^*(c_1) = \arg \max_{c_2} \mu_\pi(c_1, c_2)$ for $c_1 \in \mathcal{X}_{C_1}$. Then,

$$\leq \sum_{c_1} \mu_\pi(c_1, c_2^*(c_1)) = \sum_{y, c_1, \mathbf{x}} y P_\mathbf{x}(y, c_1) \pi(x_1|c_1, c_2^*(c_1)) \pi(x_2|c_1, c_2^*(c_1)).$$

By incorporating c_2^* into π , we can introduce π' such that $\pi(x_1|c_1, c_2^*(c_1)) \pi(x_2|c_1, c_2^*(c_1)) = \pi'(x_1|c_1) \pi'(x_2|c_1)$, satisfying Prop. 3. The variables being fixed are not necessarily conditioned on its parents (or ancestors). An example conditioning on its child is illustrated in (Fig. 5d) where we can elicit, e.g., $\pi(x_1|x_3^*(c_2), c_2) \doteq \pi'(x_1|c_2)$.

Given a general causal graph and an MPS, the aforementioned phenomena can be arbitrarily complex. We present a general criterion to test such redundancies by first proposing a lemma to obtain an intermediate expression. Let $\mathbf{V}_{\prec V}$ denote a subset of $\mathbf{V}$ preceding $V \in \mathbf{V}$ given an order $\prec$ over $\mathbf{V}$.

In separation statements involving $\pi_\mathbf{A}$, this symbol denotes independent regime nodes, each a root with an arrow into an action in $\mathbf{A}$. Each selector permits the current rule or a constant action. Such statements are evaluated in the underlying latent DAG with these roots added, so named exogenous variables are also explicit vertices. Parent and descendant sets in condition 3, and the implied closure, continue to refer to the original policy graph without selector roots.

Lemma 1. Assume finite endogenous domains and fix a deterministic optimal policy $\pi \sim \mathcal{S}$, where $\mathcal{S}$ satisfies non-redundancy (Thm. 1). Put $Q = P_\pi$ and define $K_{\mathbf{X}'}$ as in Prop. 3. Let $\mathbf{X}' \subseteq \mathbf{X}(\mathcal{S})$, actions of interest, $\mathbf{C}' \subsetneq \mathbf{C}_{\mathbf{X}'} \setminus \mathbf{X}'$, non-action contexts of interest. If there exists a subset of exogenous variables $\mathbf{U}'$ in $\mathcal{G}_\mathcal{S}$, a subset of endogenous variables $\mathbf{Z}$ in $\mathcal{G}_\mathcal{S}$ that disjoint with $\mathbf{C}' \cup \mathbf{X}'$ and subsumes $\mathbf{C}_{\mathbf{X}'} \setminus (\mathbf{C}' \cup \mathbf{X}')$, and an order $\prec$ over $\mathbf{V}' \doteq \mathbf{C}' \cup \mathbf{X}' \cup \mathbf{Z}$ such that

1. $(Y \perp\!\!\!\perp \pi_{\mathbf{X}'} \mid \lceil \mathbf{X}' \cup \mathbf{C}' \rceil)_{\mathcal{G}_\mathcal{S}}$,
2. $(C \perp\!\!\!\perp \pi_{\mathbf{X}'_{\prec C}}, \mathbf{Z}_{\prec C}, \mathbf{U}' \mid \lceil (\mathbf{X}' \cup \mathbf{C}')_{\prec C} \rceil)_{\mathcal{G}_\mathcal{S}}$ for every $C \in \mathbf{C}'$, and
3. $\mathbf{V}'_{\prec X}$ is disjoint with $de(X)_{\mathcal{G}_\mathcal{S}}$ and subsumes $pa(X)_{\mathcal{G}_\mathcal{S}}$ for every $X \in \mathbf{X}'$,

then the expected reward of π can be written as

$$\mu_\pi = \sum_{y, \mathbf{c}', \mathbf{x}'} y K_{\mathbf{x}'}(y, \mathbf{c}') \int \sum_{\mathbf{z}} \prod_{Z \in \mathbf{Z}} Q(z | \mathbf{v}'_{\prec Z}, \mathbf{u}') \prod_{X \in \mathbf{X}'} \pi(x | \mathbf{c}_x) P_{\mathbf{U}'}(d\mathbf{u}'). \quad (3)$$

Conditional distributions are normalized versions and are required to agree only on supported histories. If $Y \in \mathbf{Z}$, the auxiliary coordinate z_Y and the outer reward y are separate dummy variables.

Lemma 1 offers a sufficient condition for obtaining the intermediate expression (Eq. (3)) for us to rewrite μ_π as proposed in Prop. 3. The order $\prec$ dictates how the chain rule is applied in deriving the expression and what variables will appear as conditional for the probability terms. The first two conditions are relevant to separate $K_{\mathbf{x}'}(y, \mathbf{c}')$ from the rest. The third one is to obtain $\pi(x | \mathbf{c}_x)$ from $Q(x | \mathbf{v}'_{\prec x}, \mathbf{u}')$. We revisit Fig. 4 where we will ultimately show that, indeed C_2 and C_3 are *redundant contexts under optimality*. Given $\mathbf{C}' = \{C_1\}$ and $\mathbf{X}' = \{X_1, X_2\}$, consider $\mathbf{Z} = \{C_2, C_3\}$, $\mathbf{U}' = \emptyset$, and order $\prec = \langle C_3, C_1, X_2, C_2, X_1 \rangle$. We can derive the following expression for the expected reward (with subscripts concatenated),

$$\mu_S^* = \sum_{y, \mathbf{x}, c_1} y K_{\mathbf{x}}(y | c_1) \sum_{c_{23}} Q(c_{123}, \mathbf{x}) \quad (4)$$

$$= \sum_{y, \mathbf{x}, c_1} y K_{\mathbf{x}}(y | c_1) \sum_{c_{23}} Q(c_3) Q(c_1 | c_3) Q(x_2 | c_{13}) Q(c_2 | c_{13}, x_2) Q(x_1 | c_{123}, x_2) \quad (5)$$

$$= \sum_{c_3} Q(c_3) \sum_{y, \mathbf{x}, c_1} y K_{\mathbf{x}}(y, c_1) \sum_{c_2} Q(c_2 | c_{13}, x_2) \pi(x_2 | c_3) \pi(x_1 | c_{12}). \quad (6)$$

We now provide a sufficient condition that further polishes the intermediate expression from Lemma 1 so as to represent it as the expected reward for a smaller MPS than the original one, fulfilling the condition presented in Prop. 3.

Theorem 2. Fix the deterministic optimal policy and notation of Lemma 1, and let $\mathbf{U}'$, $\mathbf{Z}$, and $\prec$ satisfy its conditions. For $Z \in \mathbf{Z}$, choose $\mathbf{V}_Z \subseteq \mathbf{V}'_{\prec Z} \cup \mathbf{U}'$ such that normalized conditional versions satisfy $Q(Z | \mathbf{V}_Z) = Q(Z | \mathbf{V}'_{\prec Z}, \mathbf{U}')$ almost surely under Q . A minimal such set may be used but is not required. We define $\text{fixed}(\mathbf{T})$ with respect to $\{\langle Z, \mathbf{V}_Z \rangle\}_{Z \in \mathbf{Z}}$, that is, with $\hat{\mathbf{T}} \doteq [\mathbf{T}] \cup \{Z \in \mathbf{Z} \mid \mathbf{V}_Z \setminus \mathbf{U}' \subseteq [\mathbf{T}]\}$, $\text{fixed}(\mathbf{T})$ is $\mathbf{T}$ if $\mathbf{T} = \hat{\mathbf{T}}$ and $\text{fixed}(\hat{\mathbf{T}})$, otherwise. If $\text{fixed}(\mathbf{C}_X \setminus \mathbf{Z}) \supseteq \mathbf{C}_X$ for $X \in \mathbf{X}'$, then, $\mathcal{S}' \doteq (\mathcal{S} \setminus \mathbf{X}') \cup \{\langle X, \mathbf{C}_X \setminus \mathbf{Z} \rangle\}_{X \in \mathbf{X}'}$ satisfies $\mu_{\mathcal{S}'}^* = \mu_S^*$.

Thm. 2 gives a sufficient condition for removing the selected contexts without changing the optimal value. The proof fixes one common realization of $\mathbf{U}'$ and jointly chooses functions for all $Z \in \mathbf{Z}$, with expected reward at least μ_S^* . The operation fixed checks whether each original rule can then be expressed using only its retained contexts. All actions share this single realization; their rules are not optimized separately using unavailable information. The proof in Sec. E.1 also shows why arbitrary conditional versions on unreachable histories cause no difficulty.

$$\mu_S^* = \int \sum_{y, \mathbf{c}', \mathbf{x}', \mathbf{z}} y K_{\mathbf{x}'}(y, \mathbf{c}') \prod_{Z \in \mathbf{Z}} Q(z | \mathbf{v}_Z) \prod_{X \in \mathbf{X}'} \pi(x | \mathbf{c}_X) P_{\mathbf{U}'}(d\mathbf{u}'). \quad (7)$$

We explain the theorem by deriving further from Eq. (6). C_3 can be fixed to a constant c_3^* so that,

$$\leq \sum_{y, \mathbf{x}, c_1} y K_{\mathbf{x}}(y, c_1) \sum_{c_2} Q(c_2 | c_1, c_3^*, x_2) \pi(x_2 | c_3^*) \pi(x_1 | c_1, c_2). \quad (8)$$

There exists $x_2^* \in \mathfrak{X}_{X_2}$ where we can replace $\pi(x_2 | c_3^*)$ with $\pi'(x_2)$ such that $\pi'(x_2^*) = 1$.

$$\leq \sum_{y, \mathbf{x}, c_1} y K_{\mathbf{x}}(y, c_1) \sum_{c_2} Q(c_2 | c_1, c_3^*, x_2^*) \pi'(x_2) \pi(x_1 | c_1, c_2). \quad (9)$$

These steps first correspond to checking $\text{fixed}(\emptyset) = \{C_3, X_2\}$ and, then, safely replacing the decision rule for X_2 by eliminating C_3 from its context since $\text{fixed}(\mathbf{C}_{X_2} \setminus \mathbf{Z}) \supseteq \mathbf{C}_{X_2} = \{C_3\}$. Next, the optimal c_2 is determined with respect to c_1 , i.e., $Q(c_2 | c_1, c_3^*, x_2^*)$, where we can replace $\pi(x_1 | c_1, c_2^*(c_1))$ by $\pi'(x_1 | c_1)$,

$$= \sum_{c_1, c_2} Q(c_2 | c_1, c_3^*, x_2^*) \sum_{y, \mathbf{x}} y K_{\mathbf{x}}(y, c_1) \pi'(x_2) \pi(x_1 | c_1, c_2) \quad (10)$$

$$\leq \sum_{y, \mathbf{x}, c_1} y K_{\mathbf{x}}(y, c_1) \pi'(x_2) \pi(x_1 | c_1, c_2^*(c_1)) \quad (11)$$

$$= \sum_{y, \mathbf{x}, c_1} y K_{\mathbf{x}}(y, c_1) \pi'(x_1 | c_1) \pi'(x_2) = \mu_{\mathcal{S}'}^*. \quad (12)$$

These steps correspond to checking $\text{fixed}(\mathbf{C}_{X_1} \setminus \mathbf{Z}) = \text{fixed}(\{C_1\}) = \{C_1, C_3, X_2, C_2\} \supseteq \{C_1, C_2\}$ for X_1 . Since $\mu_{\mathcal{S}'}^* \leq \mu_S^*$ by the existence of $\pi \in \mathcal{S}$ that can emulate $\pi' \in \mathcal{S}'$, and $\mu_{\mathcal{S}'}^* \geq \mu_S^*$ by the derivation (Eq. (12)), we can conclude that $\mu_{\mathcal{S}'}^* = \mu_S^*$. As a consequence, MPS $\mathcal{S}$ is *not non-redundant under optimality* due to the ineffective contexts $\{C_2, C_3\}$ with respect to $\{X_1, X_2\}$.

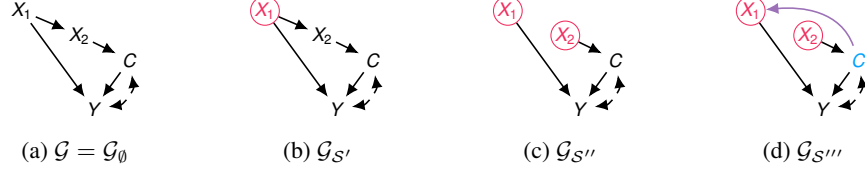


Figure 6: A causal graph $\mathcal{G}$ (a) and induced graphs (a,b,c,d). Each scope weakly improves the optimal reward of the preceding scope in every compatible model.

4 Dominance over Mixed Policies and Possible-Optimality

Non-redundancy under optimality (NRO, Def. 4) concerns comparisons with smaller scopes. We now compare scopes more generally. In Fig. 6, for example, $\mu \leq \mu_{S'}^* \leq \mu_{S''}^* \leq \mu_{S'''}^*$ in every compatible SCM. An optimal constant choice of X_1 is at least as good as its natural mechanism; after fixing X_1 , an optimal constant choice of X_2 is at least as good as its natural mechanism. Finally, allowing X_1 to observe C cannot decrease the optimum. Thus optimizing S''' suffices for the scopes displayed in this example. We call S' *weakly better* than S if $\mu_S^* \leq \mu_{S'}^*$ in every compatible SCM. To distinguish this comparison from strict possible optimality, we retain the following definition.

Definition 5 (Possibly-Optimal MPS). Given $\langle \mathcal{G}, \mathbf{X}^*, \mathbf{C}^*, Y \rangle$, let $\mathbb{S}$ be a set of NRO MPSes. An MPS $S \in \mathbb{S}$ is said to be *possibly-optimal* if there exists $\mathcal{M} \sim \mathcal{G}$ such that $\mu_S^* > \max_{S' \in \mathbb{S} \setminus \{S\}} \mu_{S'}^*$.

Unless specified otherwise, $\mathbb{S}$ contains all NRO scopes for the given tuple. Universal weak dominance is a preorder. Identifying scopes with the same optimal value in every compatible SCM gives a partial order. A POMPS cannot be weakly dominated by a distinct member of $\mathbb{S}$: that member would rule out the SCM required by the definition. We do not claim the converse. In particular, a different SCM for each competitor does not establish one SCM in which all competitors are beaten simultaneously. We next give sufficient comparisons by adding observations or interventions. A comparison excludes S from Definition 5 only when the weakly better, distinct comparator also belongs to $\mathbb{S}$.

Proposition 4. Given an MPS S and $X \in \mathbf{X}(S)$, let $C \in \mathbf{C}^* \setminus \{X\}$ and $S' = (S \setminus \{X\}) \cup \{\langle X, \mathbf{C}_X \cup \{C\} \rangle\}$. If $C \notin \text{de}(X)_{\mathcal{G}_S}$, then S' is valid and $\mu_S^* \leq \mu_{S'}^*$ in every compatible SCM.

Proof. The added edge does not create a directed cycle. Every policy for S is available in S' by ignoring C . $\square$

No independence condition is needed for this weak improvement. The enlarged scope may be redundant and therefore outside $\mathbb{S}$. For example, with $X \rightarrow Y$ and an isolated C , $\{\langle X, \emptyset \rangle\}$ can be a POMPS although $\{\langle X, \{C\} \rangle\}$ has the same optimal value in every model. The latter scope is not NRO, so it does not exclude the former.

Adding new interventions An intervention replaces a natural mechanism by a decision rule. Added contexts must provide enough information to preserve or improve its reward; this is not automatic when latent variables are shared with other mechanisms. To see a simple sufficient condition, fix a deterministic policy π optimal for S , put $Q = P_\pi$ and $\mathcal{H} = \mathcal{G}_S$, and consider $X \in \mathbf{X}^* \setminus \mathbf{X}(S)$ with new contexts $\mathbf{Z} \subseteq \mathbf{C}^* \setminus \{X\}$. If

$$(Y \perp\!\!\!\perp X \mid [\mathbf{Z}])_{\mathcal{H}_X} \quad \text{and} \quad X \notin \text{an}(\mathbf{Z})_{\mathcal{H}},$$

then the rule $\pi'(x \mid \mathbf{z}) = Q(x \mid \mathbf{z})$ reproduces the original expected reward. Here the implied variables refer to the fixed deterministic rules in π . Indeed, writing K_x for the model that retains π and fixes $X = x$ directly, the first condition gives $Q(y \mid x, \mathbf{z}) = K_x(y \mid \mathbf{z})$ on support; the second gives $K_x(\mathbf{z}) = Q(\mathbf{z})$. Thus

$$\mu_\pi = \sum_{y, x, \mathbf{z}} y K_x(y, \mathbf{z}) Q(x \mid \mathbf{z}) = \mu_{\pi \cup \{\pi'\}}.$$

Optimizing the new rule yields $\mu_S^* \leq \mu_{S \cup \{X, \mathbf{Z}\}}^*$. This single-action argument does not directly extend to simultaneous interventions: their useful contexts may be interleaved, as in Figs. 7a and 7b.

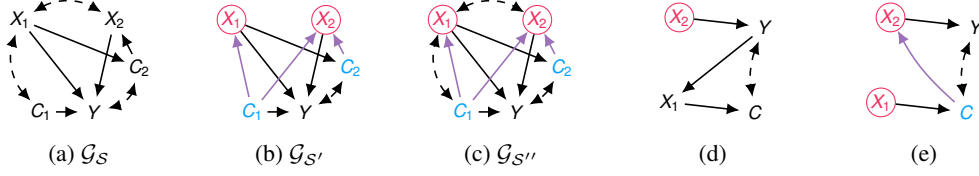


Figure 7: (a) A given scope; (b) a weakly better scope; (c) an auxiliary representation exposing information from the original latent inputs. The finite vectors of structural-function values defined below suffice for the comparison. (d,e) show how intervening after the original reward can make a context available without a cycle.

We compare two scopes through an auxiliary construction that temporarily permits otherwise unavailable information. For a fixed SCM and $X \in \mathbf{V}$, collect the values of its structural function in the vector

$$R_X = (f_X(\mathbf{p}, \mathbf{U}_X))_{\mathbf{p} \in \mathfrak{X}_{pa(X)} \mathcal{G}}.$$

Because endogenous domains are finite, R_X is finite-valued even if $\mathbf{U}_X$ is not. We retain the original exogenous variables and define R_X as their deterministic function. Shared latent inputs and their joint distribution are preserved. The vector R_X and the original endogenous parents determine the natural value of X exactly.

Theorem 3. *Let $\mathcal{S}$ and $\mathcal{S}' \neq \mathcal{S}$ be valid scopes with $\mathbf{X}(\mathcal{S}) \subseteq \mathbf{X}(\mathcal{S}')$, and suppose the directed part of $\mathcal{H}'' = \mathcal{G}_{\mathcal{S}} \cup \mathcal{G}_{\mathcal{S}'}$ is acyclic. Let*

$$\mathbf{X}' = (\mathbf{X}(\mathcal{S}') \setminus \mathbf{X}(\mathcal{S})) \cup \{X \in \mathbf{X}(\mathcal{S}) : \mathbf{C}'_X \neq \mathbf{C}_X\}.$$

For each compatible SCM, add the vectors R_X defined above and form the auxiliary scope

$$\mathcal{S}'' = (\mathcal{S} \setminus \mathbf{X}') \cup \{\langle X, pa(X)_{\mathcal{H}''} \cup \{R_X\} \rangle\}_{X \in \mathbf{X}'}.$$

Then $\mu_{\mathcal{S}}^ \leq \mu_{\mathcal{S}'}^*$ and $\mu_{\mathcal{S}'}^* \leq \mu_{\mathcal{S}''}^*$. If a sound information-removal argument establishes $\mu_{\mathcal{S}''}^* = \mu_{\mathcal{S}'}^*$, then $\mu_{\mathcal{S}}^* \leq \mu_{\mathcal{S}'}^*$ in that SCM. If the equality is established for every compatible SCM, the comparison is universal.*

The unchanged actions are retained in $\mathcal{S}''$. Newly added actions can reproduce their natural mechanisms using the corresponding R_X ; changed actions can retain their old decision rules. For each exogenous realization, the endogenous values are unchanged, proving the first inequality. The contexts of $\mathcal{S}'$ are also available in $\mathcal{S}''$, giving the second. The auxiliary scope need not respect the original $\mathbf{C}^*$; it is used only to prove a comparison between the two valid original scopes.

Thm. 2 can establish the equality when its hypotheses hold in the explicitly augmented graph, with the vectors R_X represented as deterministic children of the original latent inputs. Their dependence must not be replaced by independent copies. Appendix F gives direct arguments using one joint choice of functions for Figs. 7a to 7c and a related example. These prove weak dominance; excluding a POMPS additionally requires a distinct NRO competitor as explained above.

Refining the space of MPSes The preceding results provide sufficient tests for reducing a candidate set. A scope can be excluded from the NRO candidates when a valid proper subscope has the same optimal value in every model. Among the NRO candidates, a scope can be excluded from the POMPSes when a distinct NRO scope weakly dominates it in every model. Applying Prop. 4 alone does not suffice for this second step: the enlarged scope may be redundant. These tests do not constitute a complete characterization of POMPSes, nor by themselves establish that any particular reduced collection preserves the optimum over all scopes.

We can nevertheless remove some variables while preserving the NRO and POMPS sets exactly, provided we retain the original scope-validity constraint. Ordinary descendants of Y cannot all be discarded. In Figs. 7d and 7e, intervening on X_1 breaks the path from Y to C , making C available as a context for X_2 . The descendants that remain after cutting incoming edges to *all* eligible actions have a different status.

Proposition 5. *Given $\langle \mathcal{G}, Y, \mathbf{X}^*, \mathbf{C}^* \rangle$, let*

$$\mathbf{D} = de(Y)_{\mathcal{G}_{\mathbf{X}^*}}, \quad \mathbf{R} = \mathbf{V} \setminus \mathbf{D}, \quad \mathcal{H} = \mathcal{G}[\mathbf{R}],$$

where de excludes Y . Retain all eligible actions $\mathbf{X}^*$ and contexts $\mathbf{C}^* \setminus \mathbf{D}$, but restrict the reduced problem to

$$\mathcal{F} = \{S : \mathbf{X}(S) \subseteq \mathbf{X}^*, \mathbf{C}(S) \subseteq \mathbf{C}^* \setminus \mathbf{D}, \mathcal{G}_S \text{ is acyclic}\}.$$

The original NRO and POMPS sets are respectively equal to the NRO and POMPS sets of the reduced problem, with all proper-subscope and competitor comparisons restricted to $\mathcal{F}$. Retained domains are unchanged, and compatible mechanisms may ignore graph-allowed inputs.

Thus the smaller causal graph is accompanied by the original feasibility constraint. Checking only whether $\mathcal{H}_S$ is acyclic would allow additional scopes and can change the POMPS set; Appendix F gives a counterexample. No fixed ordering of all policies is imposed, and no additional pruning of eligible actions or retained contexts is asserted.

5 Conclusions

In this paper, we studied the space of mixed policies that emerges through the empowerment of an agent to determine the mode it will interact with the environment — i.e., which variables to intervene on and which contexts it decides to look into. Facing new challenges to optimize this new mode of interaction, which has many additional degrees of freedom, we studied the topological structure induced by the different mixed policies, which could in turn be leveraged to establish dominance comparisons across the policy space w.r.t. the maximum expected rewards achievable. As a practical result, we provided a general characterization of the space of mixed policies with respect to properties that allow the agent to detect inefficient and suboptimal strategies. One of the surprising implications of this characterization provided here is that agents following a more standard approach (i.e., intervening on all intervenable variables and observing all available contexts) may be hurting themselves, and may never be able to achieve an optimal performance regardless of the number of interactions performed.

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A Discussion – Introductory Example

We provide further elaboration on the example discussed in the introduction (Fig. 1a), which is shown again for convenience in Fig. 8. We recall that X_1 and X_2 are intervenable and X_1 and C are observable variables (i.e., can be used as context), so we write $\mathbf{X}^* = \{X_1, X_2\}$ and $\mathbf{C}^* = \{C, X_1\}$, following the corresponding notation. There are 15 distinct ways (i.e., mixed policy scopes) for an agent to interact with the system, which is explicitly shown in Fig. 9. This plot is known as a Hasse diagram (i.e., a diagram with transitive reductions) and represents the relationship between the different policies based on two dimensions:

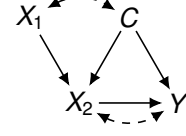


Figure 8: Introduction’s causal graph (originally, Fig. 1a).

- (a) whether one policy scope can behave always better than or equal to another with respect to maximum achievable expected rewards, which induces a dominance relationship (Fig. 9a);
- (b) whether one policy scope has more actions or contexts than another, which is called a subsumption relationship (Fig. 9b).

Regarding the dominance relation, a blue directed edge $\mathcal{S}_\alpha \rightarrow \mathcal{S}_\beta$ corresponds to $\mu_{\mathcal{S}_\alpha}^* \leq \mu_{\mathcal{S}_\beta}^*$ and a gray dotted undirected edge $\mathcal{S}_\alpha - \mathcal{S}_\beta$ represents the equivalence in their maximum achievable expected rewards, (i.e., $\mu_\alpha^* = \mu_\beta^*$); we usually call the nodes connected through these edges an *equivalence class*, given their indistinguishability in terms of achievable rewards.

Regarding the subsumption relation, a red directed edge $\mathcal{S}_\alpha \rightarrow \mathcal{S}_\beta$ represents a subsumption relationship meaning that $\mathcal{S}_\beta$ has more actions or contexts than $\mathcal{S}_\alpha$, which by itself asserts no reward ordering. With the same action set, added contexts can be ignored; adding an intervention may instead remove information carried by its natural mechanism. The goal is usually to find policies that achieve higher rewards (relative to dimension (a)) and are more parsimonious, or simpler (relative to dimension (b)). For grounding the discussion, we start with dimension (a) and consider μ , the expected reward for the observational policy, and $\mu_{\{X_1, \{C\}\}}^*$, the maximum expected reward with the policy intervening on X_1 given C . We will show below $\mu \leq \mu_{\pi(x_1|c)}^*$ by using do-calculus,

$$\mu = \sum_y yP(y) \quad \text{by definition} \quad (13)$$

$$= \sum_{y, x_1, c} yP(y, x_1, c) \quad \text{marginal probability} \quad (14)$$

$$= \sum_{y, x_1, c} yP(y|x_1, c)P(x_1|c)P(c) \quad \text{chain rule} \quad (15)$$

$$= \sum_{y, x_1, c} yP_{x_1}(y|c)P(x_1|c)P_{x_1}(c) \quad \text{Rule 2 of do-calculus} \quad (16)$$

$$= \sum_{y, x_1, c} yP_{x_1}(y, c)P(x_1|c) \quad \text{chain rule} \quad (17)$$

$$= \sum_{y, x_1, c} yP_{x_1}(y, c)\pi(x_1|c) \quad \text{by definition} \quad (18)$$

$$\leq \mu_{\{X_1, \{C\}\}}^*,$$

where the last equality comes from the expression for the expected reward (Eq. (1)) with $\pi(x_1|c)$ set to $P(x_1|c)$; the last inequality comes from the fact that the decision rule $\pi(x_1|c)$ can be optimized to yield a higher expected reward.

For further illustration of the dominance relation, we relate two policy scopes $\mathcal{S} = \{\langle X_1, \{C\} \rangle, \langle X_2, \{X_1\} \rangle\}$ and $\mathcal{S}' = \{\langle X_2, \{C\} \rangle\}$ through the following derivation,

$$\mu_{\mathcal{S}}^* = \sum_{y, x, c} yP_{\mathbf{x}}(y, c)\pi(x_1|c)\pi(x_2|x_1) \quad \text{by Eq. (1)} \quad (19)$$

$$= \sum_{y, x, c} yP_{x_2}(y, c)\pi(x_1|c)\pi(x_2|x_1) \quad \text{Rule 3 of do-calculus} \quad (20)$$

$$= \sum_{y, x_2, c} yP_{x_2}(y, c) \sum_{x_1} \pi(x_1|c)\pi(x_2|x_1) \quad \text{algebra} \quad (21)$$

There exists a probability mapping π' that can listen to C (while preserving the equality),

$$= \sum_{y, x_2, c} yP_{x_2}(y, c) \sum_{x_1} \pi(x_1|c)\pi'(x_2|x_1, c) \quad \text{by construction} \quad (22)$$

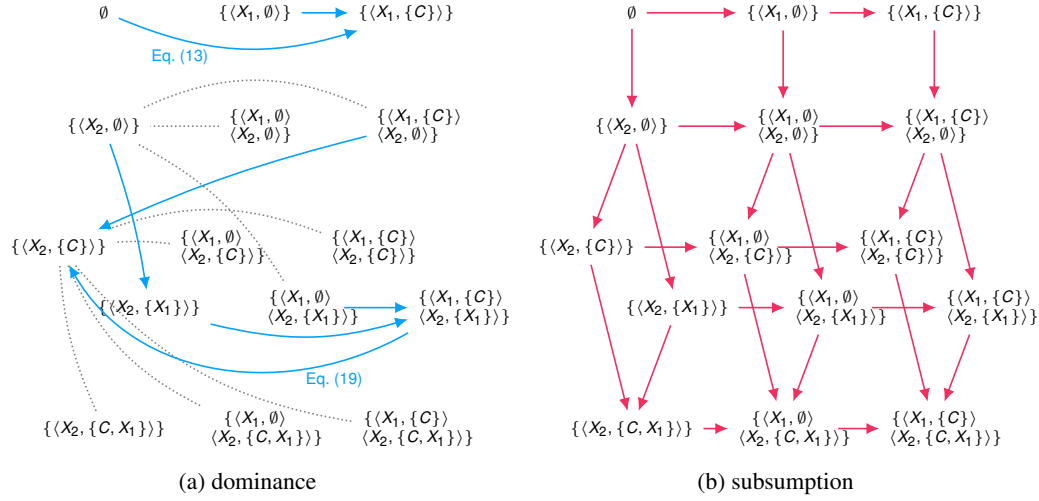


Figure 9: All 15 mixed policy scopes and their relationships in terms of two dimensions: (a) maximum achievable expected rewards and (b) policy subsumption. (a) Blue edges $\mathcal{S}_\alpha \rightarrow \mathcal{S}_\beta$ correspond to $\mu_{\mathcal{S}_\alpha}^* \leq \mu_{\mathcal{S}_\beta}^*$ with gray dotted edges the equivalence of their maximum achievable expected rewards, (b) red solid edges $\mathcal{S}_\alpha \rightarrow \mathcal{S}_\beta$ imply $\mathcal{S}_\alpha \subset \mathcal{S}_\beta$. Policy scopes are located based on interventions on X_2 (vertical) and on X_1 (horizontal). Also, their positions are preserved to facilitate corresponding comparisons.

$$= \sum_c \sum_{x_1} \pi(x_1|c) \sum_{y, x_2} y P_{x_2}(y, c) \pi'(x_2|x_1, c) \quad \text{algebra} \quad (23)$$

There exists a value x_1 for each c that can maximize the expression. Let x_1^* be a function mapping from c to such value of X_1 . Then,

$$\leq \sum_{y, x_2, c} y P_{x_2}(y, c) \pi'(x_2|x_1^*(c), c) \quad \text{by definition} \quad (24)$$

Since $x_1^*(c)$ is determined by c , there exists $\pi(x_2|c)$ such that

$$= \sum_{y, x_2, c} y P_{x_2}(y, c) \pi(x_2|c) \quad \text{by construction} \quad (25)$$

$$\leq \mu_{\mathcal{S}'}^* \quad \text{by Eq. (1).} \quad (26)$$

Therefore, $\mu_{\{\langle X_1, \{C \} \rangle, \langle X_2, \{X_1 \} \rangle\}}^* \leq \mu_{\langle X_2, \{C \} \rangle}^*$. It is not immediately obvious how we can formally derive such inequalities between two optimal expected rewards for arbitrary environments. Throughout the paper, we build graphical and algorithmic criteria that tell whether one policy can dominate another.

After all, $\{\langle X_1, \{C \} \rangle\}$ (top-right in Fig. 9a) dominates its neighbors and can attain optimality. Sometimes, a set of policies forming an equivalence class can achieve optimality, i.e.,

$$\{\langle X_2, \{C \} \rangle\}, \{\langle X_1, \emptyset \rangle, \langle X_2, \{C \} \rangle\}, \{\langle X_1, \{C \} \rangle, \langle X_2, \{C \} \rangle\}, \{\langle X_2, \{C, X_1 \} \rangle\}, \\ \{\langle X_1, \emptyset \rangle, \langle X_2, \{C, X_1 \} \rangle\}, \{\langle X_1, \{C \} \rangle, \langle X_2, \{C, X_1 \} \rangle\}.$$

Now, we turn our attention to the subsumption relation as shown in Fig. 9b. We first note that the construction of this diagram is based on the scope of the given policy (Def. 1), as defined in the paper, namely, the set of actions (before conditioning bar) and the corresponding context (after the conditioning bar). The construction is graph-insensitive, but will play a key role when combined with the analysis of dominance. In this diagram, absence of an incoming red edge within an equivalence class marks a scope minimal under subsumption in that class. The seven highlighted scopes in this example are non-redundant under optimality (NRO; Def. 4). To better understand how the dominance and subsumption dimensions are related, we superimpose both relations in Fig. 10. The scopes forming an equivalence class are clustered and highlighted within a gray rectangle, and marked its boundary with black for optimality. In this example, each equivalence class has one NRO scope, highlighted in yellow. For instance, in the aforementioned equivalence class (the right most

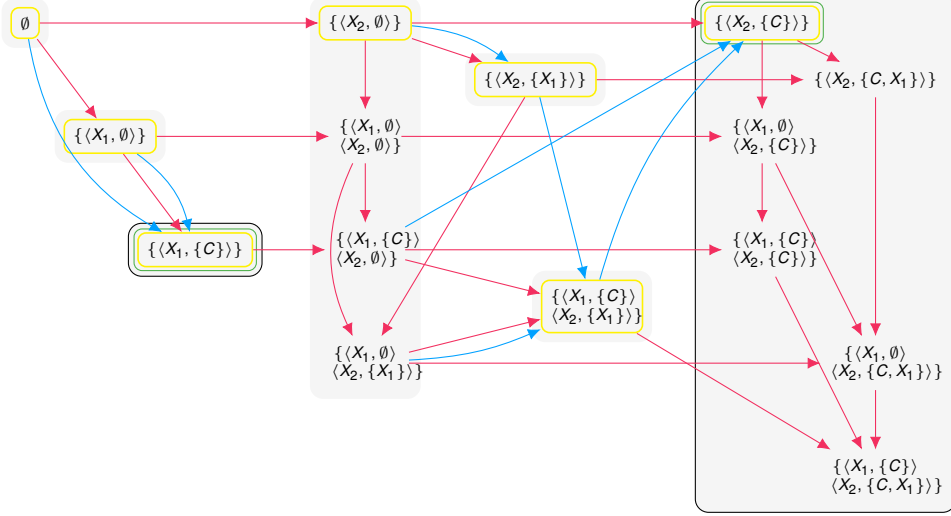


Figure 10: The superimposition of the dominance and subsumption relations of Fig. 1a, which were individually shown in Figs. 9a and 9b. The equivalence classes are highlighted in gray-shaded rectangles, policies (EC) that can achieve optimality are in black boundaries, NRO scopes are in yellow. Mixed policy scopes satisfying both NRO and optimality are in green.

equivalence class in Fig. 10), $\{\langle X_2, \{C\}\rangle\}$ is subsumed by other scopes while maintaining the same optimal reward. In fact, this implies that $\{\langle X_2, \{C\}\rangle\}$ will be preferred over its counterparts in the equivalence class given its capability of achieving the optimality while being the most parsimonious within its class. Subsumption and dominance remain distinct comparisons across equivalence classes. For example, $\{\langle X_2, \{C\}\rangle\}$ subsumes and weakly dominates $\{\langle X_2, \emptyset\rangle\}$, because the context can be ignored. It also subsumes the observational scope, but adding an intervention does not by itself imply dominance over observation. In this example, we can see through Fig. 10 that there are 7 NRO scopes (yellow): $\emptyset$, $\{\langle X_1, \emptyset\rangle\}$, $\{\langle X_1, \{C\}\rangle\}$, $\{\langle X_2, \emptyset\rangle\}$, $\{\langle X_2, \{X_1\}\rangle\}$, $\{\langle X_1, \{C\}\rangle, \langle X_2, \{X_1\}\rangle\}$, and $\{\langle X_2, \{C\}\rangle\}$. Putting this information together, we can see in Fig. 10 that only $\{\langle X_1, \{C\}\rangle\}$ and $\{\langle X_2, \{C\}\rangle\}$ satisfy both optimality (black boundaries) and NRO (yellow), which are then marked in green. These classifications concern the graph-compatible model class with freely chosen finite domains; they are not a fixed-binary completeness assertion.

Once the intelligent agent has access to causal information (e.g., in the form of the causal graph), it can explore the underlying environment with scopes that can achieve optimal reward efficiently. We now describe different approaches the agent can take. A standard approach would be taking all actions (e.g., in this case, X_1 and X_2) and observing all available contexts (X_1 and C), which leads to the scope $\{\langle X_1, \{C\}\rangle, \langle X_2, \{X_1, C\}\rangle\}$. Another approach could be brute-force, where all the 15 different scopes are experimented. A more efficient route would be to restrict the agent to the 7 NRO scopes. Knowing that 5 out of the 7 scopes are *no better than* the other 2, the most efficient approach would be to assess those, i.e., $\{\langle X_1, \{C\}\rangle\}$ and $\{\langle X_2, \{C\}\rangle\}$. We name these approaches as CB, BF, NRO, and POMPS, where the exact meaning of NRO (Non-Redundant under Optimality) and POMPS (Possibly-Optimal Mixed Policy Scope) will become clearer through out the paper.

We empirically validate that the use of refined policies leads to a better performance measured. In particular, we use the cumulative regret (the lower the better), i.e., $T\mu^* - \sum_{t=1}^T Y_t$, where T is the number of time steps (i.e., interactions) and Y_t is a random variable for the reward at time t . Further, we demonstrate that the CB approach cannot achieve the optimal reward in a certain environment, incurring a linear cumulative regret. The basic experimental setup is, for each time step, the agent assesses policies using samples from posterior reward distributions (i.e., Thompson sampling) based on its interaction history, and executes the chosen policy.³ We exemplified next an environment (structural causal model) compatible with the example discussed above, which will validate the

³Here, we mean a policy by fully-specified decision rules. For instance, there are four discrete policies corresponding to a strategy $\pi(x_1|c)$ with binary C and X_1 .

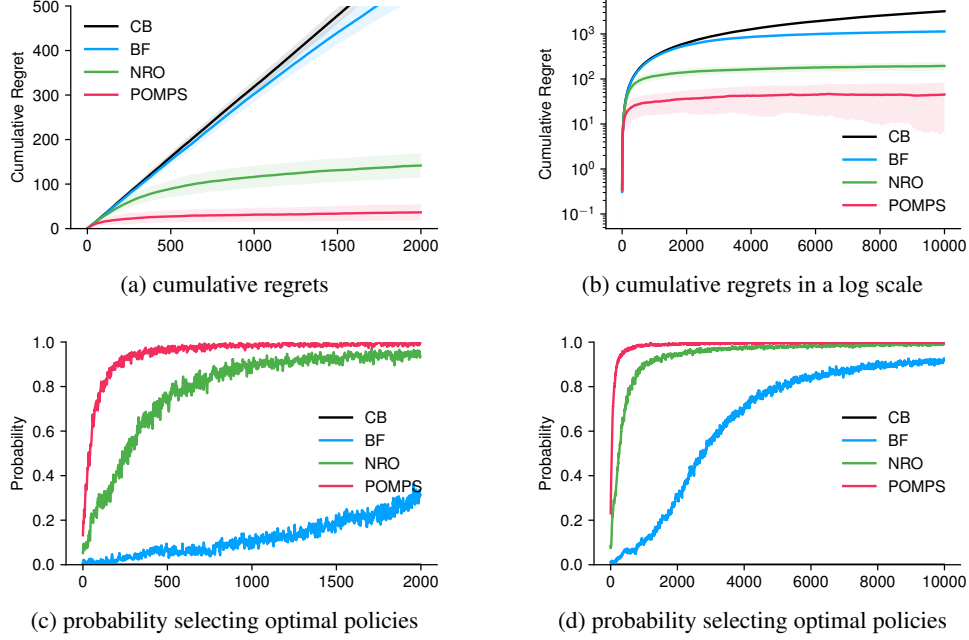


Figure 11: Performance comparison for different approaches (CB, BF, NRO, POMPS). (a,b) Each line represents cumulative regrets averaged over 100 repetitions (the lower the better), and its shade represents standard deviation. (c,d) Probability the agent selects the best policies (with CB 0% and the lines are smoothed with moving average). The figures in the left side highlight the first 2,000 time steps and ones in the right side the whole 10,000 steps.

rewards, and is unknown by the agent:

$$\mathcal{M} = \begin{cases} C \leftarrow U_C \oplus U_1 \\ X_1 \leftarrow U_1 \oplus U_{X_1} \\ X_2 \leftarrow X_1 \oplus C \oplus U_2 \oplus U_{X_2} \\ Y \leftarrow X_2 \oplus C \oplus U_2 \oplus U_Y, \end{cases}$$

where the unobserved confounders U_1 (between C and X_1) and U_2 (between X_2 and Y) are fair coins and each of $U \in \{U_{X_1}, U_{X_2}, U_C, U_Y\}$ is binary and follows $P(U = 1) = 0.1$.

The corresponding simulation is shown in Fig. 11 reporting two types of plots based on (a,b) cumulative regrets and (c,d) the probability selecting the optimal policy. It is evident from the specification that the CB agent cannot optimize its policy to achieve the optimality, regardless of the number of interactions with the environment. This is demonstrated as a linear cumulative regret (Figs. 11a and 11b) and 0% probability selecting an optimal policy in Figs. 11c and 11d. The BF approach is almost equally inefficient up to around 2000 steps, but is still able to find the optimal since it includes the possible optimal policies, POMPSes. As expected, the performance improves with the use of smaller number of policies. For this finite example, the BF, NRO, and POMPS candidate lists all contain an optimal deterministic policy. Convergence then depends on a consistent finite-arm learning procedure and its reward assumptions. The displayed simulations illustrate faster learning with the smaller lists in this setting; the theoretical scope comparisons do not imply a universal regret ordering.

The printed SCM has optimal mean $0.9^2 + 0.1^2 = 0.82$; leave X_2 natural and set $X_1 = 1$, giving $Y = 1 \oplus U_{X_2} \oplus U_Y$. Under the CB scope, the independent fair bit U_2 remains unobserved after intervening on X_2 , so its optimal mean is $1/2$. These expected values were checked exactly for this correction. The original Monte Carlo simulations and plots were not rerun.

B Preliminaries

We use in the paper classic causal inference results such as do-calculus, which we summarize here.

D-separation We start with the definition of d-separation [44], without a particular consideration of deterministic relationships.

Definition 6 (d-separation). Two sets of vertices $\mathbf{X}, \mathbf{Y}$ are said to be d-separated by another set $\mathbf{Z}$ in a directed acyclic graph $\mathcal{G}$, denoted by $(\mathbf{X} \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{Z})_{\mathcal{G}}$, if every path $\mathbf{P}$ from vertices in $\mathbf{X}$ to vertices in $\mathbf{Y}$ are blocked where blockage occurs when one of the following holds:

1. $\mathbf{P}$ contains at least one arrow-emitting node that is in $\mathbf{Z}$, or
2. $\mathbf{P}$ contains at least one collider that is outside $\mathbf{Z}$ and has no descendant in $\mathbf{Z}$.

Do-calculus Do-calculus [33] is an essential machinery to reason about the equivalence of conditional interventional probabilities induced by any model conforming to a given causal graph. Do-calculus consists of three rules where each rule ascertains that an equality between two probability distributions holds if a certain graphical test (separation) holds. The three rules are

$$\begin{aligned} \text{R1 (Insertion/deletion of observations): } & P_{\mathbf{x}}(\mathbf{y}|\mathbf{z}, \mathbf{w}) = P_{\mathbf{x}}(\mathbf{y}|\mathbf{w}) \quad \text{if } \mathbf{Z} \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{X}, \mathbf{W} \text{ in } \mathcal{G}_{\overline{\mathbf{X}}} \\ \text{R2 (Action/observation exchange): } & P_{\mathbf{x}}(\mathbf{y}|\mathbf{z}, \mathbf{w}) = P_{\mathbf{x}, \mathbf{z}}(\mathbf{y}|\mathbf{w}) \quad \text{if } \mathbf{Z} \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{X}, \mathbf{W} \text{ in } \mathcal{G}_{\overline{\mathbf{XZ}}} \\ \text{R3 (Insertion/deletion of actions): } & P_{\mathbf{x}}(\mathbf{y}|\mathbf{w}) = P_{\mathbf{x}, \mathbf{z}}(\mathbf{y}|\mathbf{w}) \quad \text{if } \mathbf{Z} \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{X}, \mathbf{W} \text{ in } \mathcal{G}_{\overline{\mathbf{XZ}(\mathbf{W})}} \end{aligned}$$

where $\mathbf{Z}(\mathbf{W})$ is a subset of $\mathbf{Z}$ that is not ancestor of $\mathbf{W}$ in $\mathcal{G}_{\overline{\mathbf{X}}}$. For convenience of some of the proofs, let $\mathcal{H} = \mathcal{G} \setminus \mathbf{X}$ and $Q = P_{\mathbf{x}}$, and consider the following equivalent rules:

$$\begin{aligned} \text{R1 (Insertion/deletion of observations): } & Q(\mathbf{y}|\mathbf{z}, \mathbf{w}) = Q(\mathbf{y}|\mathbf{w}) \quad \text{if } \mathbf{Z} \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{W} \text{ in } \mathcal{H} \\ \text{R2 (Action/observation exchange): } & Q(\mathbf{y}|\mathbf{z}, \mathbf{w}) = Q_{\mathbf{z}}(\mathbf{y}|\mathbf{w}) \quad \text{if } \mathbf{Z} \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{W} \text{ in } \mathcal{H}_{\overline{\mathbf{Z}}} \\ \text{R3 (Insertion/deletion of actions): } & Q(\mathbf{y}|\mathbf{w}) = Q_{\mathbf{z}}(\mathbf{y}|\mathbf{w}) \quad \text{if } \mathbf{Z} \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{W} \text{ in } \mathcal{H}_{\overline{\mathbf{Z}(\mathbf{W})}}. \end{aligned}$$

This representation will help us to highlight the differences between distributions while abstracting away unnecessary details.

C Mixed Policies

Derivation of Expected Reward Let $\mathbf{X} = \mathbf{X}(\pi)$ and $\mathbf{C}^- = \mathbf{C}(\pi) \setminus \mathbf{X}$. Independent policy randomizers and acyclicity give the truncated SCM factorization

$$P_{\pi}(\mathbf{v}) = \sum_{\mathbf{u}} P(\mathbf{u}) \prod_{V \notin \mathbf{X}} 1\{v = f_V(\mathbf{pa}_V, \mathbf{u}_V)\} \prod_{X \in \mathbf{X}} \pi(x|\mathbf{c}_X).$$

For fixed $(y, \mathbf{x}, \mathbf{c}^-)$, the policy product depends only on these assigned coordinates. Factoring it out and summing all remaining endogenous coordinates leaves $P_{\mathbf{x}}(y, \mathbf{c}^-)$. Multiplication by y and summation prove Eq. (1). For a subset of actions, retain all other policies and apply the same argument to the interventional distribution $K_{\mathbf{x}'}$ in Sec. 2. Sums over general exogenous variables are interpreted as integrals.

This derivation does not replace a policy-regime conditional by its natural observational counterpart. Such an exchange can fail under confounding: with a fair bit U and natural mechanisms $X = U$, $Y = U$, the policy $do(X = 0)$ gives $P_{\pi}(Y = 1|X = 0) = 1/2$, whereas $P(Y = 1|X = 0) = 0$.

Action-specific observations and multiple rewards The paper assumes a common set $\mathbf{C}^*$ of available contexts. If each action has its own available set, pure context deletion still respects availability, but context substitution and addition require separate checks. In particular, Prop. 2 does not extend without an availability condition. Let C be a fair bit, natural $A = B = 0$, and $Y = 1\{B = C\}$ in the graph $C \rightarrow Y \leftarrow B$. Suppose A may observe C , while B may observe A but not C . The scope $\{\langle A, \{C\} \rangle, \langle B, \{A\} \rangle\}$ attains one by $A = C$, $B = A$. Every scope whose actions and contexts are disjoint attains at most $1/2$: with both actions active, B has no usable input; with only B active, natural A is constant; and without an intervention on B , it remains zero. Action/context separation is valid when every input introduced by recursively substituting an upstream rule is available to the downstream action.

For multiple rewards, a specified scalar objective, such as $Y_1 + Y_2$, can be represented by an additional reward node and treated by the same framework. This fixes a particular scalar objective and is not a characterization of all multiobjective optima.

Algorithm 1 Separation of actions and contexts of an MPS

```

1: function sep-mps( $\mathcal{S}, \mathcal{G}$ )
   input: a mixed policy scope  $\mathcal{S}$ , a causal graph  $\mathcal{G}$ 
   output: an updated, action-context separated mixed policy scope  $\mathcal{S}$ 
2:   for  $X \in \text{topological-order}(\mathbf{X}(\mathcal{S}); \mathcal{G}_{\mathcal{S}})$  do
3:     Replace  $\mathbf{C}_X$  in  $\mathcal{S}$  by  $(\bigcup_{X' \in \mathbf{C}_X \cap \mathbf{X}(\mathcal{S})} \mathbf{C}_{X'}) \cup (\mathbf{C}_X \setminus \mathbf{X}(\mathcal{S}))$ .
4:   return  $\mathcal{S}$ 
  
```

D Optimality and Deterministic Mixed Policy

Proposition 1. *Given a mixed policy scope with finite endogenous domains, there exists a deterministic mixed policy, which is optimal with respect to the given scope.*

Proof. Consider any policy $\pi \sim \mathcal{S}$. Fix an action X and retain all other decision rules. Let Q'_x denote the distribution obtained by fixing $X = x$ directly, without restoring its natural mechanism. By Eq. (1),

$$\mu_{\pi} = \sum_{\mathbf{c}_X, x} \pi_X(x \mid \mathbf{c}_X) \sum_y y Q'_x(y, \mathbf{c}_X).$$

For each $\mathbf{c}_X$, choose x maximizing $\sum_y y Q'_x(y, \mathbf{c}_X)$. Replacing the probability-weighted average by this choice cannot decrease the expected reward. The other decision rules remain fixed, so Q'_x is unchanged by the replacement. A context with probability zero contributes zero for every x and permits any choice. Apply this argument to every action and context assignment to obtain a deterministic policy with reward at least μ_{π} . There are finitely many deterministic policies because the domains are finite. The best of them is therefore at least as good as every stochastic policy and is an optimal mixed policy for $\mathcal{S}$. $\square$

Proposition 2 (Separation of Actions and Contexts). *Given an MPS $\mathcal{S}$, there always exists a deterministic mixed policy $\pi \in \Pi$ such that $\mathbf{X}(\pi)$ and $\mathbf{C}(\pi)$ are disjoint and $\mu_{\pi} = \mu_{\mathcal{S}}^*$.*

Proof. Let a mixed policy $\rho \sim \mathcal{S}$ be optimal with respect to $\mathcal{S}$. First, there exists an optimal deterministic mixed-policy ρ' equivalent to ρ with respect to the expected reward (Prop. 1). Since the graph $\mathcal{G}_{\rho'}$ is acyclic, there exists a topological order among $\mathbf{X}$. Consider $X \in \mathbf{X}$ such that $\mathbf{C}_{X'} \cap \mathbf{X} = \emptyset$ for every $X' \in \mathbf{C}_X \cap \mathbf{X}$. We can create a new function π_X based on ρ'_X and $\rho'_{X'}$:

$$\pi_X((\mathbf{c}_X \setminus \{x'\}) \cup \mathbf{c}_{X'}) \doteq \rho'_X(\mathbf{C}_X \setminus \{X'\} = \mathbf{c}_x \setminus \{x'\}, X' = \rho'_{X'}(\mathbf{c}_{X'})).$$

This can be iteratively applied following the topological order among $\mathbf{X}$ to obtain a new deterministic policy π such that $\mathbf{X}(\pi)$ and $\mathbf{C}(\pi)$ are disjoint without changing the expected reward (see Alg. 1). Each replacement adds only shortcuts along directed paths of the original policy graph, so the same topological order remains valid. Every substituted context belongs to the common available set $\mathbf{C}^*$. These two facts ensure acyclicity and observability of the resulting scope. The construction preserves every endogenous value for each exogenous realization, and therefore preserves the reward. $\square$

E Non-Redundant Mixed Policy

Theorem 1. *Consider the graph-compatible nonparametric SCM class, allowing sufficiently large finite variable domains and unused structural inputs. Let $\mathcal{S} = \{\langle X, \mathbf{C}_X \rangle\}_{X \in \mathbf{X}}$ be an MPS and let $\mathcal{H} = \mathcal{G}_{\mathcal{S}}$. $\mathcal{S}$ is non-redundant if and only if (i) $\mathbf{X} \subseteq \text{an}(Y)_{\mathcal{H}}$ and (ii) $(C \not\perp\!\!\!\perp Y \mid \mathbf{C}_X \setminus \{C\})$ in $\mathcal{H} \setminus \{X\}$, for every $X \in \mathbf{X}$ and $C \in \mathbf{C}_X$.*

Proof. All subscopes in this proof are valid MPSes, so their induced graphs are acyclic. Interpret bidirected edges through a latent DAG expansion, and use independent private randomizers for policy rules. Write $Q = P_{\pi}$.

Necessity. First suppose that some action is not an ancestor of Y in $\mathcal{H}$. Let $\mathbf{W} = \text{An}(Y)_{\mathcal{H}}$ and $\mathbf{R} = \mathbf{X} \setminus \mathbf{W}$, and remove all actions in $\mathbf{R}$ together. The resulting scope $\mathcal{S}'$ is valid. Indeed, the induced graph on $\mathbf{W}$ is unchanged, and no directed edge enters $\mathbf{W}$ from its complement. Restored natural edges have their heads in $\mathbf{R}$, hence outside $\mathbf{W}$. On the complement of $\mathbf{W}$, no policy action remains, so its directed edges are a subgraph of the original acyclic $\mathcal{G}$. Neither part has a cycle, and a cycle cannot cross the two parts. The structural mechanisms determining Y are unchanged. Every policy therefore has the same reward distribution after this proper reduction. Removing just one arbitrary nonancestor action would not suffice for this argument, because restoring its natural inputs can otherwise create a cycle.

Next suppose that $C \perp\!\!\!\perp Y \mid \mathbf{K}$ in $\mathcal{H} \setminus \{X\}$, where $\mathbf{K} = \mathbf{C}_X \setminus \{C\}$. Deleting C from X 's context removes one edge and preserves validity. Define K_x directly by fixing $X = x$ and keeping all other policy rules. This model is acyclic, regardless of whether restoring the natural mechanism of X would be valid. Every variable in $\mathbf{C}_X$ is a nondescendant of X in $\mathcal{H}$, so $K_x(c, \mathbf{k}) = Q(c, \mathbf{k})$. The separation assumption gives $K_x(y \mid c, \mathbf{k}) = K_x(y \mid \mathbf{k})$ on the relevant support. Consequently,

$$\begin{aligned} Q(y) &= \sum_{x, c, \mathbf{k}} K_x(y \mid \mathbf{k}) Q(c, \mathbf{k}) \pi_X(x \mid c, \mathbf{k}) \\ &= \sum_{x, \mathbf{k}} K_x(y \mid \mathbf{k}) Q(\mathbf{k}) \underbrace{\sum_c Q(c \mid \mathbf{k}) \pi_X(x \mid c, \mathbf{k})}_{\pi'_X(x \mid \mathbf{k})}. \end{aligned}$$

The last expression is the reward distribution after the prescribed policy reduction, using the original Q . Conditional tables on null events may be completed arbitrarily. This proves necessity.

Each proper subscope. Assume both graphical conditions. Fix a proper valid subscope $\mathcal{S}' \subsetneq \mathcal{S}$, choose a topological order of $\mathcal{H}$, and let X be the latest action whose scope entry changes. Every action strictly downstream of X in $\mathcal{H}$ retains its complete context and policy table. Natural mechanisms of actions removed in $\mathcal{S}'$ are inactive under $\mathcal{S}$ and may be chosen independently of all their natural inputs. Restoring these mechanisms then introduces no functional dependence along the restored arrows.

If $\mathcal{S}'$ removes X , choose a directed path from X to Y in $\mathcal{H}$. Let the policy at X output $+1$, its natural mechanism output -1 , and each subsequent variable on this path copy its predecessor. In particular, any later action on the path has the same rule under both scopes. Ignore all other inputs. The two reward means are $+1$ and -1 . Independent sign-flip noise with a sufficiently small positive probability at every output makes all finite conditional tables strictly positive while preserving a nonzero difference.

Otherwise $\mathcal{S}'$ retains X but removes some $C \in \mathbf{C}_X$. Put $\mathbf{K} = \mathbf{C}_X \setminus \{C\}$. There is a directed path from X to Y , and an active path from C to Y in $\mathcal{H} \setminus \{X\}$ given $\mathbf{K}$. The following explicit sign construction is the single-decision supporting path construction of Everitt et al. [45, Definition 17 and Theorem 18], with the path normalization described by Everitt et al. [46, Appendix C.2, Lemma 28]. These later versions are used here as sources for the corrected proof.

Truncate the active path at its first intersection V with the directed X -to- Y path, and follow the latter path from V to Y . The active path arrives at V with an arrowhead. Otherwise, tracing its last collider-free section backwards from V reaches either C or a collider with a directed path to $\mathbf{K}$. This would make a context of X a descendant of X , contrary to acyclicity. Each collider on the active path has an activation path to $\mathbf{K}$; stop that path at its first vertex in $\mathbf{K}$. These supporting paths can be normalized so that intersecting activation paths are replaced by the route through their intersection, reducing the number of colliders. Where distinct transmitted quantities share a variable, assign separate coordinates of that variable to them.

Partition the active path into its collider-free sections. At their sources generate independent fair signs $S_0, \dots, S_m$. Copy the sign of each section along its directed edges. Thus C receives S_0 , and the final section delivers S_m to V . At collider j , form the product $S_{j-1}S_j$ and copy it along its activation path to a context in $\mathbf{K}$. A context reached by several activation paths carries the corresponding vector of products. Every other context coordinate is constant. In particular, no coordinate in $\mathbf{K}$ carries an individual source sign: an interior noncollider in $\mathbf{K}$ would block the active path, and an activation wire transmits only its product coordinate. The directed X -to- Y path cannot pass through any context of X .

Let the distinguished coordinate of the X -rule be

$$D_X = S_0 \prod_{j=1}^m (S_{j-1} S_j) = S_m.$$

Transmit D_X along the directed path to V , multiply it there by the incoming sign S_m , and copy this product to Y . The reward is therefore $Y = +1$ under the full policy. All other active actions on the supporting paths are treated as stochastic kernels with precisely their allowed $\mathcal{H}$ -parents; separate coordinates implement any shared paths.

To handle the paper's prescribed marginalization and simultaneous changes at earlier actions, we need more than a single-decision value comparison. Flip all source signs S_j simultaneously. Source and copy coordinates are odd under this involution, collider products and activation coordinates are even, and every coordinate in $\mathbf{K}$ is unchanged. The decision coordinate D_X is odd, while the product at V is even. Give every transmitted coordinate an independent sign-flip noise of probability $\epsilon \in (0, 1/2)$, including decision and reward outputs. Give unused coordinates independent full-support noise. Keep these noise variables fixed under the involution. The same parity relations still hold, and all observable conditional tables now have full support. Hence, exactly and for every $\mathbf{k}$,

$$Q(D_X = +1 \mid \mathbf{k}) = Q(D_X = -1 \mid \mathbf{k}) = \frac{1}{2}.$$

If the reduced context is $\mathbf{K}' \subseteq \mathbf{K}$, averaging again gives the same identity conditional on $\mathbf{K}'$. Since X is intervened on with an independent private policy randomizer, its prescribed reduced rule is

$$\pi'_X(x \mid \mathbf{k}') = \sum_{\mathbf{c}_X \setminus \mathbf{k}'} \pi_X(x \mid \mathbf{c}_X) Q(\mathbf{c}_X \setminus \mathbf{k}' \mid \mathbf{k}') = Q(x \mid \mathbf{k}').$$

Its distinguished output is therefore a fresh fair sign. If X has extra coordinates, choose them as independent fair signs as well, so they do not reveal or constrain this output.

Under the reduced policy the reward has the form $Y = D_X B E$, where B is the sign arriving on the side path and E is the product of the independent noises on the directed reward path. Earlier retained actions can lose context inputs, and earlier removed actions revert to their chosen input-independent natural mechanisms. Neither operation creates a functional route from the fresh D_X into the side-path coordinates. In particular, side-path coordinates at shared variables ignore the decision-path coordinate. Rules at all downstream actions are unchanged by the choice of the latest changed action. Thus D_X remains independent of (B, E) and the reduced mean is zero. The full-policy mean tends to one as ϵ tends to zero: if N noisy operations affect the test, the probability of any flip is at most $N\epsilon$, and the mean is at least $1 - 2N\epsilon > 0$ for sufficiently small ϵ . This gives a finite SCM and policy separating $\mathcal{S}$ from the particular simultaneous reduction $\mathcal{S}'$, with strictly positive entries in the finite conditional probability tables just constructed.

One SCM and one policy. There are finitely many proper valid subsopes $\mathcal{S}'_1, \dots, \mathcal{S}'_r$. Embed the constructions above in common finite vector domains large enough for all their coordinates; unused coordinates can be independent noise. Choose finite latent domains large enough for all shared-source coordinates. The reward can always have domain $\{-1, +1\}$. Parameterize all natural conditional tables, latent root distributions, and full-policy tables by independent simplex coordinates θ , restricted to their interiors. These stochastic tables are realized by structural equations with independent private response-function noises in addition to the graph's shared latent roots. They describe a nonempty relatively open parameter space of compatible SCMs and policies.

For each i , the original law P_π is polynomial in θ . Every reduced rule is rational in θ , because it averages using conditional probabilities of this same original law. All denominators are strictly positive. The acyclic factorization for $\mathcal{S}'_i$ therefore makes

$$\Delta_i(\theta) = \mu_\pi(\theta) - \mu_{\pi'_i}(\theta) = \frac{p_i(\theta)}{q_i(\theta)}, \quad q_i(\theta) > 0,$$

a rational function in the simplex coordinates. The construction for $\mathcal{S}'_i$ shows that p_i is not the zero polynomial. Their finite product is consequently nonzero and cannot vanish on an open set. Choose a positive parameter point outside its zero set. This supplies one SCM and one policy for which every Δ_i is nonzero, as required by Def. 3. No assertion that these differences have the same sign is required or made. If $r = 0$, the requirement is vacuous. $\square$

Algorithm 2 Non-Redundant Mixed Policy Scope $\mathcal{S}$

```
1: function NR-MPS( $\mathcal{G}, Y, \mathcal{S}$ )  
   input: a mixed policy scope  $\mathcal{S}$ , a causal graph  $\mathcal{G}$   
   output: an updated, non-redundant mixed policy scope  $\mathcal{S}$   
2:    $\mathcal{S} \leftarrow \mathcal{S} \setminus (\mathbf{X}(\mathcal{S}) \setminus \text{an}(Y)_{\mathcal{G}_{\mathcal{S}}})$ .  
3:   for  $X \in \text{reverse-order}(\mathbf{X}(\mathcal{S}); \mathcal{G}_{\mathcal{S}})$  do  
4:     if  $X \notin \text{an}(Y)_{\mathcal{G}_{\mathcal{S}}}$  then:  
5:        $\mathcal{S} \leftarrow \mathcal{S} \setminus \{X\}$  and continue.  
6:     for  $C \in \mathbf{C}_X$  do  
7:       if  $(C \perp\!\!\!\perp Y \mid \mathbf{C}_X \setminus \{C\})$  in  $\mathcal{G}_{\mathcal{S}} \setminus \{X\}$  then  
8:          $\mathcal{S} \leftarrow (\mathcal{S} \setminus \{X\}) \cup \{X, \mathbf{C}_X \setminus \{C\}\}$ .  
9:   return  $\mathcal{S}$ .
```

Pruning a scope. A *graphical reduction* of $\mathcal{S}$ is a scope obtained by finitely many of the following operations on the current scope $\mathcal{U}$: (i) delete $C \in \mathbf{C}_X$ when $C \perp\!\!\!\perp Y \mid \mathbf{C}_X \setminus \{C\}$ in $\mathcal{G}_{\mathcal{U}} \setminus \{X\}$; (ii) remove a nonempty set of active actions outside $\text{An}(Y)_{\mathcal{G}_{\mathcal{U}}}$, provided the resulting scope is valid. Removing all such actions together is always valid, by the necessity proof of Thm. 1. A graphical reduction is *minimal* if neither operation applies. A related formulation of unique minimal policy-space reductions appears in Zhang and Bareinboim [47, Definitions 5–6 and Theorem 1]; validity of action removal is made explicit here for general MPSes.

We retain the original one-pass procedure. After its initial bulk deletion, fix any topological order of the remaining actions and scan it in reverse. At each action, scan its current contexts once, using the current induced graph for every test. Removing an action removes its scope entry, while retaining uses of the now-natural variable as another action’s context.

Proposition 6. *Given a valid MPS, every intermediate scope in Alg. 2 is valid.*

Proof. The initial bulk deletion is valid by the ancestral-set argument in the necessity proof of Thm. 1. Let $\mathcal{H}_0$ be the graph immediately after it, with the fixed topological action order. Context deletion only removes an edge. Inductively, subsequent operations create no new directed reachability between remaining actions, and each already processed, retained action still reaches Y . To see the latter, a directed path from such an action to Y can enter only actions later in the fixed order; their incoming edges are no longer changed.

When the current action X fails the ancestor test, it has no other remaining action as a descendant: a processed one would give a path to Y , and an unprocessed one would violate the fixed order and the reachability invariant. Its proper descendants thus all have natural mechanisms. Restoring the natural inputs of X cannot create a cycle: a resulting cycle through X would consist entirely of original natural edges and would be a cycle in $\mathcal{G}$. It cannot create a new path between remaining actions, since such a path would have to leave X and reach another action. This establishes the induction and validity. $\square$

Theorem 4. *Given a valid MPS $\mathcal{S}$, Alg. 2 returns its unique minimal graphical reduction $\mathcal{S}^{\text{out}}$, independently of the permitted action and context processing orders. In fact, $\mathcal{S}^{\text{out}} \subseteq \mathcal{T}$ for every graphical reduction $\mathcal{T}$ of $\mathcal{S}$. The output satisfies both conditions of Thm. 1 and is non-redundant in that theorem’s model class. For every compatible SCM and policy $\pi \sim \mathcal{S}$, every prescribed reduction $\pi^{\text{out}} \subseteq \pi$ on $\mathcal{S}^{\text{out}}$ has the same reward distribution as π .*

Proof. The finite scans terminate, and validity follows from the preceding proposition. Its invariant also shows that every retained action remains an ancestor of Y . We prove that every context retained at its test continues to satisfy the second condition of Thm. 1.

A graphical observation. Use a latent DAG expansion for bidirected edges. In a current graph $\mathcal{H}$, let $\mathbf{A} = \text{An}(Y)_{\mathcal{H}}$ and let K be the undirected moral graph of $\mathcal{H}[\mathbf{A}]$: drop arrowheads and join every pair of parents of a common child. For an action $Z \in \mathbf{A}$ with policy parents $\mathbf{P}_Z$, all its parents are contexts and

$$\text{An}(\mathbf{P}_Z \cup \{Y\})_{\mathcal{H} \setminus \{Z\}} = \mathbf{A} \setminus \{Z\}.$$

For $D \in \mathbf{P}_Z$, the separation test in the algorithm is therefore equivalent to separating D from Y in $K \setminus \mathbf{B}$, where $\mathbf{B} = \{Z\} \cup (\mathbf{P}_Z \setminus \{D\})$. Indeed, the extra moral edges created by Z join its parents, and all but D are conditioned on. The set $\mathbf{B}$ is a clique.

Suppose this test deletes $D \rightarrow Z$. Let W be the connected component of D in $K \setminus \mathbf{B}$, so $Y \notin W$. Let K' be the reward-ancestral moral graph after the deletion. Every vertex that loses its ancestry of Y belongs to W : its former directed path to Y must use $D \rightarrow Z$, and the prefix to D avoids $\mathbf{B}$, whose vertices still reach Y . Among retained vertices, removing $D \rightarrow Z$ can remove only its own undirected edge and moral edges from D to the other parents of Z . A lost ancestor can also remove a moral edge between its parents. If both parents are outside W , both lie in $\mathbf{B}$, since every neighbor of W outside W lies there; their edge survives because $\mathbf{B}$ remains a clique at Z . Consequently

$$K'[\mathbf{A} \setminus W] = K[\mathbf{A} \setminus W].$$

For any conditioning set E and two endpoints outside $W \cup E$, every connecting path in $K \setminus E$ can thus be replaced by a path in $K' \setminus E$: shortcut each excursion into W through the surviving clique $\mathbf{B} \setminus E$.

Preservation of a retained context. Fix a context C retained for action X . First consider deleting another context D of the same action. In the observation above, $Z = X$, so $C \in \mathbf{B}$ and $C \notin W$. Use the old conditioning/deletion set $E = \{X\} \cup (\mathbf{P}_X \setminus \{C\})$. The old connection survives in $K' \setminus E$; releasing the now deleted context D from E cannot destroy it. Hence contexts already retained remain dependent after the rest of X 's context scan.

Next consider a later context deletion at another action Z . The fixed reverse order and reachability invariant imply that X is not an ancestor of Z . A directed path $X \rightarrow \dots \rightarrow Y$ therefore avoids $\mathbf{B} = \{Z\} \cup (\mathbf{P}_Z \setminus \{D\})$. If $C \notin \mathbf{B}$, the path $C \rightarrow X \rightarrow \dots \rightarrow Y$ also avoids $\mathbf{B}$; otherwise C is in the clique itself. In either case $C \notin W$, and $Y \notin W$. Apply the graphical observation with $E = \{X\} \cup (\mathbf{P}_X \setminus \{C\})$ to preserve the context's connection to Y .

Finally, deleting a nonancestor action only restores inputs to a vertex outside $An(Y)$, leaving the reward-ancestral induced graph and its moral graph unchanged. Thus it cannot invalidate any of these retained connections. Both graphical conditions hold at the end, establishing NR in the model class of Thm. 1.

Uniqueness and minimality. Consider the graphical reduction operations at any current scope $\mathcal{U}$. Write $\mathcal{H}_{\mathcal{U}} = \mathcal{G}_{\mathcal{U}}$ and $\mathbf{A}_{\mathcal{U}} = An(Y)_{\mathcal{H}_{\mathcal{U}}}$. Every valid removal of nonancestor actions leaves the induced graph on $\mathbf{A}_{\mathcal{U}}$ and the ancestral set itself unchanged: restored edges have their heads outside that set, and no edge enters it from outside. Let $B(\mathcal{U})$ remove all such actions together. Any valid partial removal can be followed by bulk removal to reach $B(\mathcal{U})$.

Every step decreases the number of action and context entries. We show that any two available steps have a common continuation. Context deletions at different actions commute, since deleting an edge preserves d-separation with fixed endpoints and conditioning set. For two deletions at the same action X , write its contexts as $\mathbf{K} \cup \{C, D\}$. In the same graph $\mathcal{H}_{\mathcal{U}} \setminus \{X\}$, the two tests give

$$C \perp\!\!\!\perp Y \mid \mathbf{K}, D, \quad D \perp\!\!\!\perp Y \mid \mathbf{K}, C \quad \implies \quad \{C, D\} \perp\!\!\!\perp Y \mid \mathbf{K}.$$

This is graphical intersection: in the common ancestral moral graph, a path from Y to the first of C, D encountered after deleting $\mathbf{K}$ would contradict one of the two tests. Thus either deletion leaves the other available. This graph property requires no positivity assumption on the SCM.

Two action-removal branches both continue to $B(\mathcal{U})$. For an action removal and a context deletion $C \rightarrow X$, if $X \notin \mathbf{A}_{\mathcal{U}}$, bulk removal absorbs the context deletion and both branches reach $B(\mathcal{U})$. Otherwise, X , its contexts, and every ancestor needed for its separation test belong to $\mathbf{A}_{\mathcal{U}}$. Action removal leaves that test unchanged. Perform the context deletion on both branches and then remove all remaining nonancestor actions. The branches agree: the context deletion gives the same smaller reward-ancestral induced graph, and restorations outside the old ancestral set cannot create a path into it.

Termination and these joining cases imply, by induction on the number of scope entries, that every exhaustive pruning sequence ends at the same scope $\mathcal{R}$. Every graphical reduction $\mathcal{T}$ of $\mathcal{S}$ can continue to $\mathcal{R}$, so $\mathcal{R} \subseteq \mathcal{T}$. The terminal scope is therefore the unique minimal, indeed least, graphical reduction. Terminality is exactly the pair of graphical conditions in Thm. 1. Every step of Alg. 2

is a valid graphical reduction, and the preceding argument shows that its output is terminal. Thus $\mathcal{S}^{\text{out}} = \mathcal{R}$ for every permitted processing order.

Preservation under the prescribed policy reduction. Write $Q = P_\pi$ for the original policy law. At each context deletion, average the current policy using the current context law. The fixed-action calculation in the necessity proof of Thm. 1 preserves the reward distribution. Removing nonancestor actions does so as well. We show that their composition agrees with the prescribed reduction using Q .

Immediately before an action X is processed, its joint law with its original contexts $\mathbf{P}_X$ is still $Q(X, \mathbf{P}_X)$. The initial bulk deletion preserves this law. Subsequent earlier steps in the reverse scan change only downstream or incomparable actions; an action restored to its natural mechanism has no remaining action as a descendant, by the preceding validity proof. These changes cannot affect X or its contexts. During X 's context scan, its own rule cannot affect its contexts. Successive averaging therefore uses the same original context law. If $\mathbf{D}_X$ are its final contexts, the tower property gives

$$\pi_X^{\text{out}}(x \mid \mathbf{D}_X) = E_Q[\pi_X(x \mid \mathbf{P}_X) \mid \mathbf{D}_X] = Q(x \mid \mathbf{D}_X) \quad Q\text{-almost surely.}$$

If no context is removed, the original rule is retained pointwise.

We also verify that later pruning cannot activate an original-null assignment of $\mathbf{D}_X$. Consider a later deletion $D \rightarrow B$, with other current contexts $\mathbf{Z}$, so $D \perp\!\!\!\perp Y \mid \mathbf{Z}$ in $\mathcal{H} \setminus \{B\}$. The reverse order gives $X \notin \text{An}(B)_\mathcal{H}$. For every $V \in \mathbf{D}_X \setminus (\mathbf{Z} \cup \{B\})$, an active path from D to V given $\mathbf{Z}$ would concatenate with $V \rightarrow X \rightarrow \dots \rightarrow Y$ to contradict this separation. Indeed, the active path has vertices only among ancestors of $D, V, \mathbf{Z}$, so it contains neither X nor a descendant of X ; the directed continuation avoids B and $\mathbf{Z}$. The case $D \in \mathbf{D}_X$ is itself excluded by the direct path $D \rightarrow X \rightarrow \dots \rightarrow Y$. Graphical composition consequently gives

$$D \perp\!\!\!\perp \{Y\} \cup (\mathbf{D}_X \setminus (\mathbf{Z} \cup \{B\})) \mid \mathbf{Z} \quad \text{in } \mathcal{H} \setminus \{B\}.$$

The same fixed-action calculation, now with joint target $(Y, \mathbf{D}_X)$, proves that averaging at B preserves this joint law. If $B \in \mathbf{D}_X$, keep $B = b$ in that calculation instead of summing it out. Removing a nonancestor action likewise preserves $\mathbf{D}_X$, since its members are reward ancestors. Hence the final $\mathbf{D}_X$ -law equals its original Q -law.

The sequential construction thus agrees with the prescribed kernels on their original supports, and those supports remain unchanged. Changing a kernel at a null parent assignment leaves the joint law of an acyclic model unchanged. Apply this fact successively to all retained actions to obtain every permitted original-null completion. All prescribed reductions to the returned scope therefore preserve the original reward distribution. No positivity assumption is used. $\square$

Minimality and uniqueness here concern graphical reductions of the input scope. They do not assert maximality among all NR subscores or uniqueness among all semantically reward-preserving subscores. No positivity, perfect-recall, or solubility assumption is added.

E.1 Non-Redundancy under Optimality

Derivations for the Redundancy of Examples We demonstrate the redundancy for Figs. 5b to 5d. We may employ ' $\leq$ ' to highlight that fixing operation can improve the expected reward although, given the optimality of the left hand side, it becomes '='. We present a derivation for Fig. 5b showing that C_3 is non-informative.

$$\mu_\pi = \sum_{y, \mathbf{x}, \mathbf{c}} yQ(y \mid \mathbf{x}, \mathbf{c})Q(\mathbf{x}, \mathbf{c}) \quad \text{MP,CR (27)}$$

$$= \sum_{y, \mathbf{x}, \mathbf{c}} yQ(y \mid \mathbf{x}, c_1, c_2)Q(\mathbf{x}, \mathbf{c}) \quad \text{R1 (28)}$$

$$= \sum_{y, \mathbf{x}, \mathbf{c}} yQ'_\mathbf{x}(y \mid c_1, c_2)Q(\mathbf{x}, \mathbf{c}) \quad \text{R1,R2 (29)}$$

$$= \sum_{y, \mathbf{x}, \mathbf{c}} yQ'_\mathbf{x}(y \mid c_1, c_2)Q(x_1 \mid c_3, c_1)Q(x_2 \mid c_3, c_2)Q(\mathbf{c}) \quad \text{CR,R1 (30)}$$

$$= \sum_{y, \mathbf{x}, c_1, c_2} yQ'_\mathbf{x}(y \mid c_1, c_2)Q(c_1, c_2) \sum_{c_3} Q(x_1 \mid c_3, c_1)Q(x_2 \mid c_3, c_2)Q(c_3) \quad \text{MP,CR (31)}$$

$$= \sum_{c_3} Q(c_3) \sum_{y, \mathbf{x}, c_1, c_2} y Q'_x(y|c_1, c_2) Q(c_1, c_2) Q(x_1|c_3, c_1) Q(x_2|c_3, c_2) \quad \text{algebra (32)}$$

$$\leq \sum_{y, \mathbf{x}, c_1, c_2} y Q'_x(y|c_1, c_2) Q(c_1, c_2) Q(x_1|c_3^*, c_1) Q(x_2|c_3^*, c_2) \quad \text{def (33)}$$

$$= \sum_{y, \mathbf{x}, c_1, c_2} y Q'_x(y|c_1, c_2) Q(c_1, c_2) \pi'(x_1|c_1) \pi'(x_2|c_2) \quad \text{def (34)}$$

$$= \sum_{y, \mathbf{x}, c_1, c_2} y Q'_x(y|c_1, c_2) Q'_x(c_1, c_2) \pi'(x_1|c_1) \pi'(x_2|c_2) \quad \text{R1,R3 (35)}$$

$$= \sum_{y, \mathbf{x}, c_1, c_2} y Q'_x(y, c_1, c_2) \pi'(x_1|c_1) \pi'(x_2|c_2) \quad \text{CR (36)}$$

where $c_3^* \in \mathcal{X}_{C_3}$ is the value maximizing the inner sum.

The derivation for Fig. 5c is given that C_2 is non-informative.

$$\mu_\pi = \sum_{y, \mathbf{x}, \mathbf{c}} y Q(y|\mathbf{x}, \mathbf{c}) Q(\mathbf{x}, \mathbf{c}) \quad \text{def (37)}$$

$$= \sum_{y, \mathbf{x}, \mathbf{c}} y Q(y|\mathbf{x}, c_1) Q(\mathbf{x}, \mathbf{c}) \quad \text{R1 (38)}$$

$$= \sum_{y, \mathbf{x}, \mathbf{c}} y Q'_x(y|c_1) Q(\mathbf{x}, \mathbf{c}) \quad \text{R1,R2 (39)}$$

$$= \sum_{y, \mathbf{x}, \mathbf{c}} y Q'_x(y|c_1) Q(c_1) Q(c_2|c_1) Q(x_1|\mathbf{c}) Q(x_2|\mathbf{c}) \quad \text{CR (40)}$$

$$= \sum_{y, \mathbf{x}, \mathbf{c}} y Q'_x(y|c_1) Q'_x(c_1) Q(c_2|c_1) Q(x_1|\mathbf{c}) Q(x_2|\mathbf{c}) \quad \text{R1,R3 (41)}$$

$$= \sum_{y, \mathbf{x}, \mathbf{c}} y Q'_x(y, c_1) Q(c_2|c_1) Q(x_1|\mathbf{c}) Q(x_2|\mathbf{c}) \quad \text{CR (42)}$$

$$= \sum_{y, \mathbf{x}, c_1} y Q'_x(y, c_1) \sum_{c_2} Q(c_2|c_1) Q(x_1|\mathbf{c}) Q(x_2|\mathbf{c}) \quad \text{algebra (43)}$$

$$= \sum_{c_1} \sum_{c_2} Q(c_2|c_1) \sum_{y, \mathbf{x}} y Q'_x(y, c_1) Q(x_1|\mathbf{c}) Q(x_2|\mathbf{c}) \quad \text{algebra (44)}$$

$$\leq \sum_{c_1} \sum_{y, \mathbf{x}} y Q'_x(y, c_1) Q(x_1|c_1, c_2^*(c_1)) Q(x_2|c_1, c_2^*(c_1)) \quad \text{def (45)}$$

$$= \sum_{y, \mathbf{x}, c_1} y Q'_x(y, c_1) \pi'(x_1|c_1) \pi'(x_2|c_1) \quad \text{def (46)}$$

The derivation for Fig. 5d is as follows. Let $Q'_{\mathbf{x}_{12}}$ denote the distribution obtained by fixing X_1, X_2 and retaining all other decision rules.

$$\mu_\pi = \sum_{y, \mathbf{x}_{12}, c_2} y Q(y|\mathbf{x}_{12}, c_2) Q(\mathbf{x}_{12}, c_2) \quad \text{MP,CR (47)}$$

$$= \sum_{y, \mathbf{x}_{12}, c_2} y Q'_{\mathbf{x}_{12}}(y|c_2) Q(\mathbf{x}_{12}, c_2) \quad \text{R1,R2 (48)}$$

$$= \sum_{y, \mathbf{x}_{12}, c_2} y Q'_{\mathbf{x}_{12}}(y|c_2) \sum_{x_3} Q(\mathbf{x}, c_2) \quad \text{MP (49)}$$

$$= \sum_{y, \mathbf{x}_{12}, c_2} y Q'_{\mathbf{x}_{12}}(y|c_2) \sum_{x_3} Q(c_2) Q(x_3|c_2) Q(x_1|x_3, c_2) Q(x_2|x_3, c_2, x_1) \quad \text{CR (50)}$$

$$= \sum_{y, \mathbf{x}_{12}, c_2} y Q'_{\mathbf{x}_{12}}(y|c_2) Q'_{\mathbf{x}_{12}}(c_2) \sum_{x_3} Q(x_3|c_2) Q(x_1|x_3, c_2) Q(x_2|x_3, c_2) \quad \text{R2,R3 (51)}$$

$$= \sum_{y, \mathbf{x}_{12}, c_2} y Q'_{\mathbf{x}_{12}}(y, c_2) \sum_{x_3} Q(x_3|c_2) Q(x_1|x_3, c_2) Q(x_2|x_3, c_2) \quad \text{CR (52)}$$

$$= \sum_{c_2} \sum_{x_3} Q(x_3|c_2) \sum_{y, \mathbf{x}_{12}} y Q'_{\mathbf{x}_{12}}(y, c_2) Q(x_1|x_3, c_2) Q(x_2|x_3, c_2) \quad \text{algebra (53)}$$

$$\leq \sum_{y, \mathbf{x}_{12}, c_2} y Q'_{\mathbf{x}_{12}}(y, c_2) Q(x_1|x_3^*(c_2), c_2) Q(x_2|x_3^*(c_2), c_2) \quad \text{def (54)}$$

$$= \sum_{y, \mathbf{x}_{12}, c_2} y Q'_{\mathbf{x}_{12}}(y, c_2) \pi'(x_1|c_2) \pi'(x_2|c_2) \quad \text{def (55)}$$

Proofs and Additional Characterizations

Lemma 1. Assume finite endogenous domains and fix a deterministic optimal policy $\pi \sim \mathcal{S}$, where $\mathcal{S}$ satisfies non-redundancy (Thm. 1). Put $Q = P_\pi$ and define $K_{\mathbf{x}'}$ as in Prop. 3. Let $\mathbf{X}' \subseteq \mathbf{X}(\mathcal{S})$, actions of interest, $\mathbf{C}' \subseteq \mathbf{C}_{\mathbf{X}'} \setminus \mathbf{X}'$, non-action contexts of interest. If there exists a subset of exogenous variables $\mathbf{U}'$ in $\mathcal{G}_{\mathcal{S}}$, a subset of endogenous variables $\mathbf{Z}$ in $\mathcal{G}_{\mathcal{S}}$ that disjoint with $\mathbf{C}' \cup \mathbf{X}'$ and subsumes $\mathbf{C}_{\mathbf{X}'} \setminus (\mathbf{C}' \cup \mathbf{X}')$, and an order $\prec$ over $\mathbf{V}' \doteq \mathbf{C}' \cup \mathbf{X}' \cup \mathbf{Z}$ such that

1. $(Y \perp\!\!\!\perp \pi_{\mathbf{X}'} \mid [\mathbf{X}' \cup \mathbf{C}'])_{\mathcal{G}_{\mathcal{S}}}$,
2. $(C \perp\!\!\!\perp \pi_{\mathbf{X}' \prec C}, \mathbf{Z}_{\prec C}, \mathbf{U}' \mid [(\mathbf{X}' \cup \mathbf{C}')_{\prec C}])_{\mathcal{G}_{\mathcal{S}}}$ for every $C \in \mathbf{C}'$, and
3. $\mathbf{V}'_{\prec X}$ is disjoint with $\text{de}(X)_{\mathcal{G}_{\mathcal{S}}}$ and subsumes $\text{pa}(X)_{\mathcal{G}_{\mathcal{S}}}$ for every $X \in \mathbf{X}'$,

then the expected reward of π can be written as

$$\mu_\pi = \sum_{y, \mathbf{c}', \mathbf{x}'} y K_{\mathbf{x}'}(y, \mathbf{c}') \int \sum_{\mathbf{z}} \prod_{Z \in \mathbf{Z}} Q(z|\mathbf{v}'_{\prec Z}, \mathbf{u}') \prod_{X \in \mathbf{X}'} \pi(x|\mathbf{c}_x) P_{\mathbf{U}'}(d\mathbf{u}'). \quad (3)$$

Conditional distributions are normalized versions and are required to agree only on supported histories. If $Y \in \mathbf{Z}$, the auxiliary coordinate z_Y and the outer reward y are separate dummy variables.

Proof. Write $\mathbf{A} = \mathbf{X}'$, $\mathbf{C} = \mathbf{C}'$, $\mathbf{L} = \mathbf{U}'$, and $\mathbf{W} = \mathbf{V}'$. Let $\mathbf{B} = \mathbf{X}(\mathcal{S}) \setminus \mathbf{A}$, and write the fixed deterministic policy as $f = (f_X)_{X \in \mathbf{X}(\mathcal{S})}$, with law $Q = P_f$. For each $\mathbf{a}$, define $K_{\mathbf{a}}$ directly by retaining $f_{\mathbf{B}}$ and assigning $\mathbf{A} = \mathbf{a}$. Its graph is $\mathcal{G}_{\mathcal{S}}$ with incoming arrows at $\mathbf{A}$ removed. In particular, no intermediate model obtained by restoring natural action mechanisms is needed.

For $C \in \mathbf{C}$, choose normalized conditional distributions

$$b_C(c \mid \mathbf{a}_{\prec C}, \mathbf{c}_{\prec C})$$

from the interventional distributions $K_{\mathbf{a}}$. By condition 3, C and the preceding variables in $\mathbf{C}$ are nondescendants of every action in $\mathbf{A} \setminus \mathbf{A}_{\prec C}$. Their joint distribution under these interventions therefore does not depend on those later action values. These conditional distributions, including their versions at null conditioning events, can consequently be chosen by the displayed action prefix, so that

$$K_{\mathbf{a}}(\mathbf{c}) = \prod_{C \in \mathbf{C}} b_C(c \mid \mathbf{a}_{\prec C}, \mathbf{c}_{\prec C}). \quad (56)$$

Also choose a normalized version $r_{\mathbf{a}}(y \mid \mathbf{c})$ of $K_{\mathbf{a}}(y \mid \mathbf{c})$.

We first justify two conditional-probability identities on histories supported by Q :

$$Q(y \mid \mathbf{a}, \mathbf{c}) = r_{\mathbf{a}}(y \mid \mathbf{c}), \quad (57)$$

$$Q(c \mid \mathbf{w}_{\prec C}, \mathbf{l}) = b_C(c \mid \mathbf{a}_{\prec C}, \mathbf{c}_{\prec C}). \quad (58)$$

For Eq. (58), put $\mathbf{I} = \mathbf{A}_{\prec C}$, $\mathbf{S}_C = (\mathbf{A} \cup \mathbf{C})_{\prec C}$, and $\mathbf{D}_C = [\mathbf{S}_C]$. Introduce independent finite regime nodes only at the actions in $\mathbf{I}$, each choosing either its original rule or a constant action. Every action whose rule changes is already conditioned on in $\mathbf{S}_C$. Each additional variable in $\mathbf{D}_C$ is therefore the same deterministic function of $\mathbf{S}_C$ in every regime. In particular, any later selected action inferred by this closure still uses its original rule in this comparison.

Condition 2 implies independence of C from these selectors, $\mathbf{Z}_{\prec C}$, and $\mathbf{L}$ given $\mathbf{D}_C$. Since the closure adds only deterministic functions, it gives the corresponding conditional-probability identity conditional on $\mathbf{S}_C$. Compare the original regime with the regime fixing $\mathbf{I}$ to its observed values. An original supported history is possible after these interventions: replacing each of these rules by its realized output preserves that realization. Thus the original conditional of C equals its conditional

under $do(\mathbf{I} = \mathbf{a}_\mathbf{I})$ given the preceding $\mathbf{C}$ variables. Only now fix the remaining selected actions. By condition 3, these later actions cannot affect C or the preceding $\mathbf{C}$ variables. This yields Eq. (58). The same argument proves Eq. (57) using condition 1: introduce selectors at all actions in $\mathbf{A}$, all of which are already conditioned on, and compare their original rules with their constant values. No equality with the natural observational conditional is asserted.

For each $Z \in \mathbf{Z}$, let $k_Z(z \mid \mathbf{w}_{\prec Z}, \mathbf{l})$ be a normalized version of $Q(z \mid \mathbf{w}_{\prec Z}, \mathbf{l})$. Condition 3 puts every context of an action $X \in \mathbf{A}$ before X in $\prec$, and hence its conditional distribution is $1\{x = f_X(\mathbf{c}_X)\}$. The chain rule and Eqs. (56) and (58) give the full marginal identity

$$Q(\mathbf{a}, \mathbf{c}, \mathbf{z}, d\mathbf{l}) = K_{\mathbf{a}}(\mathbf{c})P_{\mathbf{L}}(d\mathbf{l}) \prod_{Z \in \mathbf{Z}} k_Z(z \mid \mathbf{w}_{\prec Z}, \mathbf{l}) \cdot \prod_{X \in \mathbf{A}} 1\{x = f_X(\mathbf{c}_X)\}. \quad (59)$$

This is an equality of measures in $\mathbf{l}$; no finite or discrete exogenous domain is assumed. The endogenous variables are finite-valued, so their conditional distributions are specified by conditional probabilities.

To verify Eq. (59) without a positivity assumption, replace the chain-rule factors in the sequential order $\prec$. At every reached history the replacement agrees with the original factor. The next prefix thus has the original law, so it cannot introduce a previously null history. Induction gives the stated identity with arbitrary normalized versions at unreachable histories. The same argument permits the reduced conditional distributions of Thm. 2, when their stipulated almost-sure equalities hold.

Define an auxiliary joint law by

$$T(y, \mathbf{a}, \mathbf{c}, \mathbf{z}, d\mathbf{l}) = r_{\mathbf{a}}(y \mid \mathbf{c})Q(\mathbf{a}, \mathbf{c}, \mathbf{z}, d\mathbf{l}). \quad (60)$$

Its $(\mathbf{A}, \mathbf{C}, \mathbf{Z}, \mathbf{L})$ marginal is the corresponding marginal of Q , and its expected reward is μ_f by Eq. (57). If $Y \in \mathbf{Z}$, the coordinate z_Y in this marginal and the outer coordinate y in T are separate dummy copies; the construction does not assert that their original joint dependence is retained. Substituting Eq. (59) into Eq. (60) and using $r_{\mathbf{a}}(y \mid \mathbf{c})K_{\mathbf{a}}(\mathbf{c}) = K_{\mathbf{a}}(y, \mathbf{c})$ proves Eq. (3). $\square$

Theorem 2. Fix the deterministic optimal policy and notation of Lemma 1, and let $\mathbf{U}'$, $\mathbf{Z}$, and $\prec$ satisfy its conditions. For $Z \in \mathbf{Z}$, choose $\mathbf{V}_Z \subseteq \mathbf{V}'_{\prec Z} \cup \mathbf{U}'$ such that normalized conditional versions satisfy $Q(Z \mid \mathbf{V}_Z) = Q(Z \mid \mathbf{V}'_{\prec Z}, \mathbf{U}')$ almost surely under Q . A minimal such set may be used but is not required. We define $\text{fixed}(\mathbf{T})$ with respect to $\{\langle Z, \mathbf{V}_Z \rangle\}_{Z \in \mathbf{Z}}$, that is, with $\hat{\mathbf{T}} \doteq [\mathbf{T}] \cup \{Z \in \mathbf{Z} \mid \mathbf{V}_Z \setminus \mathbf{U}' \subseteq [\mathbf{T}]\}$, $\text{fixed}(\mathbf{T})$ is $\mathbf{T}$ if $\mathbf{T} = \hat{\mathbf{T}}$ and $\text{fixed}(\hat{\mathbf{T}})$, otherwise. If $\text{fixed}(\mathbf{C}_X \setminus \mathbf{Z}) \supseteq \mathbf{C}_X$ for $X \in \mathbf{X}'$, then, $\mathcal{S}' \doteq (\mathcal{S} \setminus \mathbf{X}') \cup \{\langle X, \mathbf{C}_X \setminus \mathbf{Z} \rangle\}_{X \in \mathbf{X}'}$ satisfies $\mu_{\mathcal{S}'}^* = \mu_{\mathcal{S}}^*$.

Proof. Use the notation and conditional distributions from the preceding proof, replacing k_Z by the stipulated normalized version $k_Z(z \mid \mathbf{v}_Z)$. Write $\mathbf{R}_X = \mathbf{C}_X \setminus \mathbf{Z}$. We construct one policy depending only on these retained contexts.

First sample $\mathbf{l}$ with law $P_{\mathbf{L}}$. Conditional on this draw, choose a function t_Z for each $Z \in \mathbf{Z}$ by sampling its output at every possible assignment of $\mathbf{V}_Z \setminus \mathbf{L}$ from $k_Z(\cdot \mid \mathbf{v}_Z)$, using the corresponding coordinates of the same $\mathbf{l}$. These outputs are sampled independently across variables and input assignments, conditional on $\mathbf{l}$. There are finitely many such assignments because the endogenous domains are finite. Let $\omega = (\mathbf{l}, (t_Z)_{Z \in \mathbf{Z}})$ and denote its law by ν .

For fixed ω , define T_ω by replacing the exogenous draw and each k_Z in the factorization of T by their selected value and deterministic function, respectively. Keep the context distributions b_C , action rules f_X , and final reward distribution $r_{\mathbf{a}}$ unchanged. This is a normalized sequential distribution in the order $\prec$: a context distribution uses earlier actions and contexts, a function t_Z uses earlier $\mathbf{W}$ coordinates, and an action uses its earlier contexts. For each fixed assignment of $\mathbf{W}$, each t_Z is evaluated once. The independence of the sampled outputs therefore gives

$$T = \int T_\omega \nu(d\omega), \quad \mu_f = \int J(\omega) \nu(d\omega), \quad J(\omega) = \mathbb{E}_{T_\omega}[y]. \quad (61)$$

We must select the realization while also preserving support. There are only finitely many assignments of $\mathbf{W}$. Since the mixture's $\mathbf{W}$ marginal is $Q(\mathbf{W})$, every Q -null assignment has zero probability under

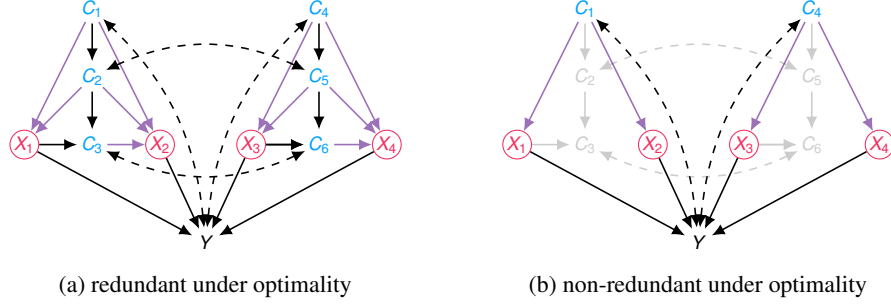


Figure 12: A more involved example of redundancy under optimality in a mixed policy scope

T_ω for almost every ω . Removing a finite union of null sets gives

$$\text{supp } T_\omega(\mathbf{W}) \subseteq \text{supp } Q(\mathbf{W}) \quad \text{for } \nu\text{-almost every } \omega. \quad (62)$$

Choose one ω^* in this full-measure set with $J(\omega^*) \geq \mu_f$. Such a choice exists by Eq. (61); a bounded random variable cannot be strictly below its mean on a full-measure set. This fixes one common realization for all actions and context values, without assuming an attained maximum over exogenous values.

For each $X \in \mathbf{A}$, run the fixed construction from $\mathbf{R}_X$, attaching a function of $\mathbf{R}_X$ to each inferred value. An implied action J is assigned its original function f_J of its inferred contexts. An inferred Z is assigned its selected function t_Z evaluated at its inferred inputs; the exogenous coordinates are already fixed by ω^* . The hypothesis says that this closure recovers every original context of X , and therefore expresses f_X as a function $g_X(\mathbf{R}_X)$.

These decision rules agree with the original actions on the selected law's support. To see this, induct through the finite closure construction. The equalities defining t_Z hold by the definition of T_{ω^*} . Deterministic policy equalities among $\mathbf{W}$ coordinates hold by Eq. (62). If an implied action lies outside $\mathbf{W}$, compose its unchanged function with the already inferred inputs. When that action is subsequently used to infer a $\mathbf{W}$ coordinate, the resulting composed equality is an equality among $\mathbf{W}$ coordinates under Q , and is therefore inherited by the selected law. This covers actions in $\mathbf{B}$ as well as those in $\mathbf{A}$, without identifying the entire outside-variable distribution with Q . Choose any fixed derivation when closure offers several; they agree on the selected support. Define the resulting rules arbitrarily on retained-context assignments absent from that support. We obtain

$$g_X(\mathbf{r}_X) = f_X(\mathbf{c}_X) \quad T_{\omega^*}\text{-almost surely, for every } X \in \mathbf{A}.$$

Replace all the f_X by g_X in the sequential generator of T_{ω^*} . Its law remains unchanged. Indeed, by induction through $\prec$, the reached prefix has the old law, and the replacement action agrees with the old action on every reached history. Thus arbitrary off-support values never create new histories. This establishes simultaneous substitution, rather than assuming that an almost-sure identity survives an arbitrary change of policy.

Now $\mathbf{R}_X \subseteq \mathbf{A} \cup \mathbf{C}$ for every selected action. Consequently no context distribution, resulting action rule, or final reward distribution depends on $\mathbf{Z}$ or $\mathbf{L}$. Summing the normalized sequential factors for $\mathbf{Z}$ gives one for each fixed $(\mathbf{a}, \mathbf{c})$. Hence

$$J(\omega^*) = \sum_{y, \mathbf{a}, \mathbf{c}} y K_{\mathbf{a}}(y, \mathbf{c}) \prod_{X \in \mathbf{A}} 1\{x = g_X(\mathbf{r}_X)\}. \quad (63)$$

By the expected-reward formula, this is precisely the expected reward of the actual SCM policy retaining $f_{\mathbf{B}}$ and using $g_{\mathbf{A}}$. Its scope is S' . It is valid because deleting context arrows while retaining all interventions leaves a subgraph of the acyclic $\mathcal{G}_S$. Thus $\mu_{S'}^* \geq J(\omega^*) \geq \mu_f = \mu_S^*$. The reverse inequality follows because S can implement every S' policy by ignoring the deleted contexts. This proves the asserted equality. $\square$

The four-action example. In Fig. 12, fix a deterministic optimal policy with rules $f_1(c_1, c_2)$, $f_2(c_1, c_2, c_3)$, $f_3(c_4, c_5)$, and $f_4(c_4, c_5, c_6)$. Let $\mathbf{U}'$ collect all exogenous inputs to C_2, C_3, C_5, C_6 ,

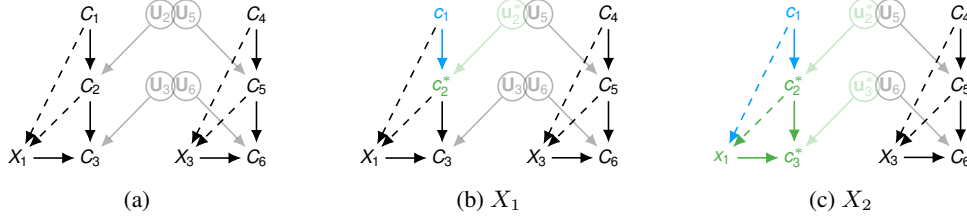


Figure 13: (a) Dependencies of the auxiliary contexts and the deterministic action rules (dashed arrows). (b) Given C_1 , the common selected auxiliary realization determines C_2 and hence X_1 . (c) The same realization and inferred X_1 determine C_3 , allowing the second action to use only C_1 . The other two actions are treated symmetrically using C_4 . The starred auxiliary inputs are components of one jointly selected realization.

including their private noises and shared inputs. Keep their joint distribution. In the displayed graph these inputs are independent of the exogenous inputs governing C_1, C_4, Y ; fixing the four actions removes their remaining paths to the reward. For one joint realization ω of $\mathbf{U}'$, write the natural context mechanisms as

$$c_2 = h_2(c_1, \omega), \quad c_3 = h_3(c_2, x_1, \omega), \quad c_5 = h_5(c_4, \omega), \quad c_6 = h_6(c_5, x_3, \omega).$$

Define four rules using only the retained contexts:

$$\begin{aligned} g_1^\omega(c_1) &= f_1(c_1, h_2(c_1, \omega)), \\ g_2^\omega(c_1) &= f_2(c_1, h_2(c_1, \omega), h_3(h_2(c_1, \omega), g_1^\omega(c_1), \omega)), \\ g_3^\omega(c_4) &= f_3(c_4, h_5(c_4, \omega)), \\ g_4^\omega(c_4) &= f_4(c_4, h_5(c_4, \omega), h_6(h_5(c_4, \omega), g_3^\omega(c_4), \omega)). \end{aligned}$$

For almost every ω , the reward under these decision rules has the law of the original policy reward conditional on $\mathbf{U}' = \omega$. The reward-side mechanisms and their exogenous distribution are unchanged, and the action values agree by construction. Consequently

$$\mu_f = \int \mu_{g^\omega} P_{\mathbf{U}'}(d\omega).$$

There exists a single ω^* with $\mu_{g^{\omega^*}} \geq \mu_f$. All four rules share this fixed realization; they do not choose auxiliary inputs separately for different observed contexts. The reduced scope is valid because only context arrows are removed. It achieves at least the original optimum, and the reverse inequality follows by ignoring contexts. Thus its optimal value is unchanged. No maximizer over exogenous values or movement of context-dependent conditional probabilities outside their summations is required. Fig. 13 illustrates the deterministic substitutions.

Remark 1. $\mathcal{S}' = \{\langle X_1, \{C_1\} \rangle, \langle X_2, \{C_1\} \rangle, \langle X_3, \{C_4\} \rangle, \langle X_4, \{C_4\} \rangle\}$ is non-redundant under optimality.

Proof. Let the independent confounders shared by (C_1, Y) and (C_4, Y) be $U_1 = (L_1, L_2)$ and $U_4 = (L_3, L_4)$, respectively, with four independent fair bits. Set $C_1 = U_1$, $C_4 = U_4$, take binary actions with natural mechanisms $X_i = 0$, and ignore other inputs. Define

$$Y = 1\{(X_1, X_2, X_3, X_4) = (L_1, L_2, L_3, L_4)\}.$$

The full scope attains one by returning the designated bit at each action. Every proper subscope either removes an action, leaving its natural value zero, or removes its sole context. That action then matches its independent fair target bit with probability at most $1/2$. Since all four matches are required, every proper valid subscope has optimal reward at most $1/2$. Thus this single SCM establishes NRO. A reward checking only the parity of the eight bits would not establish this claim: one action could transmit both bits' parity while the other action remained constant. $\square$

The argument also allows additional shared inputs among C_2, C_3, C_5, C_6 : retain them jointly in $\mathbf{U}'$. The required independence from the reward-side inputs must still hold; arbitrary new confounding with C_1, C_4 , or Y is not covered.

F Dominance over Mixed Policies and Possible-Optimality

Theorem 3. Let $\mathcal{S}$ and $\mathcal{S}' \neq \mathcal{S}$ be valid scopes with $\mathbf{X}(\mathcal{S}) \subseteq \mathbf{X}(\mathcal{S}')$, and suppose the directed part of $\mathcal{H}'' = \mathcal{G}_{\mathcal{S}} \cup \mathcal{G}_{\mathcal{S}'}$ is acyclic. Let

$$\mathbf{X}' = (\mathbf{X}(\mathcal{S}') \setminus \mathbf{X}(\mathcal{S})) \cup \{X \in \mathbf{X}(\mathcal{S}) : \mathbf{C}'_X \neq \mathbf{C}_X\}.$$

For each compatible SCM, add the vectors R_X defined above and form the auxiliary scope

$$\mathcal{S}'' = (\mathcal{S} \setminus \mathbf{X}') \cup \{(X, pa(X)_{\mathcal{H}''} \cup \{R_X\})\}_{X \in \mathbf{X}'}$$

Then $\mu_{\mathcal{S}}^* \leq \mu_{\mathcal{S}''}^*$ and $\mu_{\mathcal{S}'}^* \leq \mu_{\mathcal{S}''}^*$. If a sound information-removal argument establishes $\mu_{\mathcal{S}''}^* = \mu_{\mathcal{S}'}^*$, then $\mu_{\mathcal{S}}^* \leq \mu_{\mathcal{S}'}^*$ in that SCM. If the equality is established for every compatible SCM, the comparison is universal.

Proof. Fix a compatible SCM. Add R_X as a deterministic function of the original $\mathbf{U}_X$ for each $X \in \mathbf{X}'$. By the definition of R_X , $f_X(\mathbf{p}, \mathbf{u}_X) = R_X(\mathbf{p})$; this does not change any potential outcome. In particular, if two mechanisms share an exogenous variable, their vectors R_X depend on that same variable. They are not independent copies. The vectors R_X have no endogenous parents in the original graph, and their introduction does not create a directed cycle.

Take a deterministic policy π for $\mathcal{S}$. In the auxiliary scope, retain its decision rules at every old action, including the unchanged actions in $\mathcal{S} \setminus \mathbf{X}'$. The contexts of a changed old action are included in $pa(X)_{\mathcal{H}''}$. At a newly intervened action, evaluate R_X at its original endogenous parent values. Those parents are present in $\mathcal{H}''$ because this action is not intervened on in $\mathcal{S}$. Induction along a topological order of the union graph shows that every original endogenous variable, and hence Y , has exactly its original value for every exogenous state. Thus $\mu_{\mathcal{S}}^* \leq \mu_{\mathcal{S}''}^*$.

Every rule of $\mathcal{S}'$ is also available in $\mathcal{S}''$, by ignoring any extra inputs, so $\mu_{\mathcal{S}'}^* \leq \mu_{\mathcal{S}''}^*$. All augmented endogenous domains are finite, so these optima are attained. The asserted conclusion follows whenever $\mu_{\mathcal{S}''}^* = \mu_{\mathcal{S}'}^*$ is established. Repeating the argument for every compatible SCM gives the universal comparison. An application of Thm. 2 must verify its hypotheses in this augmented model, retaining the deterministic definition of the vectors R_X and all shared latent inputs. $\square$

The simultaneous-intervention example. Consider Figs. 7a to 7c. Let R_1, R_2 be the vectors of structural-function values for X_1, X_2 , respectively, and write $K_{\mathbf{x}}(y, c_1, c_2) = P_{\mathbf{x}}(y, c_1, c_2)$. In the auxiliary scope, a deterministic policy has the form

$$x_1 = g_1(c_1, r_1), \quad x_2 = g_2(c_1, c_2, r_2).$$

The graph gives the following expected reward, retaining the dependence between R_1 and R_2 :

$$\mu_g = \sum_{y, \mathbf{x}, c_1, c_2, r_1, r_2} y K_{\mathbf{x}}(y, c_1, c_2) q(r_1, r_2 \mid c_1) 1\{x_1 = g_1(c_1, r_1)\} 1\{x_2 = g_2(c_1, c_2, r_2)\}.$$

To justify the factorization, fixing the actions removes the latent inputs to their natural mechanisms. The latent inputs shared with C_1 are separated from the remaining reward mechanism by C_1 ; those shared only by the actions have no remaining effect. The inputs shared by C_2 and Y remain together inside $K_{\mathbf{x}}$. Thus $q(r_1, r_2 \mid c_1)$ is independent of the reward-side variables conditional on c_1 and the fixed actions. Conditional values at zero-probability c_1 can be chosen arbitrarily.

Independently for every possible c_1 , draw a pair $(r_1(c_1), r_2(c_1))$ from $q(\cdot, \cdot \mid c_1)$, thereby choosing a random function T of c_1 . For a fixed realization t , define

$$g_1^t(c_1) = g_1(c_1, r_1^t(c_1)), \quad g_2^t(c_1, c_2) = g_2(c_1, c_2, r_2^t(c_1)).$$

Averaging the expected reward of these policies over T gives exactly μ_g . Therefore some *single* function t has reward at least μ_g . The two rules then use only the contexts in $\mathcal{S}'$; the function is a fixed part of the policy, not an additional observation. This proves $\mu_{\mathcal{S}''}^* \leq \mu_{\mathcal{S}'}^*$, and the reverse inequality follows by ignoring the auxiliary inputs. The construction applies to arbitrary original exogenous domains because only the finite-valued R_1, R_2 are sampled.

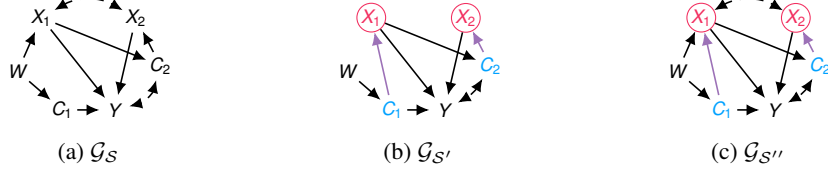


Figure 14: A variant of Figs. 7a to 7c: a non-contextualizable W affects X_1 and C_1 instead of their sharing a latent input. The target scope lets X_1 observe C_1 and X_2 observe C_2 . The last panel schematically retains the additional information used in the auxiliary comparison.

The variant with an unavailable observed parent. In Fig. 14, W is not contextualizable. The vectors (R_1, R_2) are independent of (W, C_1) , although R_1 and R_2 may be dependent. An auxiliary policy has rules $x_1 = g_1(c_1, w, r_1)$ and $x_2 = g_2(c_2, r_2)$, with expected reward

$$\mu_g = \sum_{y, \mathbf{x}, c_1, c_2, w, r_1, r_2} y K_{\mathbf{x}}(y, c_1, c_2) q(w | c_1) q(r_1, r_2) 1\{x_1 = g_1(c_1, w, r_1)\} 1\{x_2 = g_2(c_2, r_2)\}.$$

Draw one joint pair (r_1, r_2) and one function $w(c_1)$ whose values are sampled independently from $q(\cdot | c_1)$. Averaging the resulting policies reproduces μ_g , so some single choice (r_1^*, r_2^*, w^*) has at least that reward. Its rules are

$$g_1'(c_1) = g_1(c_1, w^*(c_1), r_1^*), \quad g_2'(c_2) = g_2(c_2, r_2^*).$$

This proves the comparison in the displayed example; in particular, X_2 does not need to observe C_1 . It does not assert that all results extend without additional conditions to arbitrary action-specific observation sets.

Proposition 5. Given $\langle \mathcal{G}, Y, \mathbf{X}^*, \mathbf{C}^* \rangle$, let

$$\mathbf{D} = de(Y)_{\mathcal{G}_{\overline{\mathbf{X}^*}}}, \quad \mathbf{R} = \mathbf{V} \setminus \mathbf{D}, \quad \mathcal{H} = \mathcal{G}[\mathbf{R}],$$

where de excludes Y . Retain all eligible actions $\mathbf{X}^*$ and contexts $\mathbf{C}^* \setminus \mathbf{D}$, but restrict the reduced problem to

$$\mathcal{F} = \{\mathcal{S} : \mathbf{X}(\mathcal{S}) \subseteq \mathbf{X}^*, \mathbf{C}(\mathcal{S}) \subseteq \mathbf{C}^* \setminus \mathbf{D}, \mathcal{G}_{\mathcal{S}} \text{ is acyclic}\}.$$

The original NRO and POMPS sets are respectively equal to the NRO and POMPS sets of the reduced problem, with all proper-subscope and competitor comparisons restricted to $\mathcal{F}$. Retained domains are unchanged, and compatible mechanisms may ignore graph-allowed inputs.

Proof. We prove both that all original NRO scopes belong to $\mathcal{F}$ and that one model transformation preserves all their reward comparisons simultaneously.

For each $d \in \mathbf{D}$, there is a directed path from Y to d in $\mathcal{G}_{\overline{\mathbf{X}^*}}$. This path contains no eligible action. Hence $\mathbf{D} \cap \mathbf{X}^* = \emptyset$, and the path remains present in $\mathcal{G}_{\mathcal{S}}$ for every originally valid scope $\mathcal{S}$. Every $d \in \mathbf{D}$ is therefore a strict descendant of Y in $\mathcal{G}_{\mathcal{S}}$ and cannot be an ancestor of Y .

Suppose a valid scope uses $d \in \mathbf{D}$ as a context of an action X . Then $Y \rightarrow \dots \rightarrow d \rightarrow X$ remains present, so acyclicity implies that X is not an ancestor of Y . Removing the single context d from the rule for X preserves validity and does not affect any reward ancestor. Every policy can consequently be replaced by a policy for this proper subscope with the same reward, and the reverse inequality follows by ignoring d . Thus this scope is not NRO. Every original NRO scope uses only contexts in $\mathbf{C}^* \setminus \mathbf{D}$ and belongs to $\mathcal{F}$.

Now fix an original SCM $\mathcal{M}$ and choose one value $\mathbf{d}_0 \in \mathfrak{X}_{\mathbf{D}}$. For each retained variable $V \in \mathbf{R}$, define

$$f_V^R(\mathbf{pa}_R, \mathbf{u}_V) = f_V(\mathbf{pa}_R, \mathbf{d}_0|_{pa(V)_{\mathcal{G} \cap \mathbf{D}}}, \mathbf{u}_V).$$

Keep the original joint distribution of all latent inputs needed by the retained mechanisms. The resulting SCM $\mathcal{M}_R$ is compatible with the induced subgraph $\mathcal{H}$. The construction is the same for every scope. For any $\mathcal{S} \in \mathcal{F}$, no variable in $\mathbf{D}$ is an ancestor of Y in $\mathcal{G}_{\mathcal{S}}$. Therefore the displayed substitution leaves the reward-ancestral mechanisms unchanged. Coupling the same exogenous state and the same independent policy randomness gives

$$Y_{\mathcal{M}}^{\pi} = Y_{\mathcal{M}_R}^{\pi} \quad \text{for every } \mathcal{S} \in \mathcal{F} \text{ and every } \pi \sim \mathcal{S}.$$

In particular, $\mu_S^*(\mathcal{M}) = \mu_S^*(\mathcal{M}_R)$ simultaneously for all $S \in \mathcal{F}$.

Conversely, extend any $\mathcal{H}$ -compatible SCM to $\mathcal{G}$ by supplying mechanisms for the deleted variables and making each retained mechanism ignore its additional deleted-parent inputs. Preserve the joint latent inputs to the retained mechanisms; graph-allowed extra inputs may also be ignored. This is permitted by our compatibility convention. The same policywise reward equalities hold, without changing any retained domain. Consequently, the two model classes realize exactly the same vectors $(\mu_S^*)_{S \in \mathcal{F}}$.

Every originally valid proper subscope of a member of $\mathcal{F}$ also belongs to $\mathcal{F}$. The value-vector equality therefore preserves its NRO status in both directions. We already proved that there are no original NRO scopes outside $\mathcal{F}$. Finally, the NRO competitor sets coincide, and each SCM satisfying the strict POMPS inequalities gives one of the same simultaneously attainable value vectors. The POMPS sets are thus equal as well. $\square$

Why the original validity constraint is necessary. Consider the graph

$$A \rightarrow Y \rightarrow Z \rightarrow X \rightarrow C, \quad X \leftrightarrow Y,$$

with eligible actions $\{A, X\}$ and contexts $\{C\}$. The original reduction rule in Proposition 5 deletes Z and retains both actions and C . The remaining induced graph is $A \rightarrow Y, X \rightarrow C, X \leftrightarrow Y$. In that smaller graph, let U be a fair binary latent variable and define

$$A = 0, \quad X = U, \quad C = X, \quad Y = 1\{A = U\}.$$

All endogenous variables are binary. The scope $S = \{\langle A, \{C\} \rangle\}$ achieves reward 1 by setting $A = C$. Every other valid scope has optimum 1/2: without intervening on X , an action A not observing C cannot predict U ; intervening on X removes the only information about U carried by C . A policy for X cannot observe C because $X \rightarrow C \rightarrow X$ would be a cycle. Thus S is NRO and a strict POMPS in the unrestricted smaller-graph problem.

In the original graph, however, this scope creates the cycle $A \rightarrow Y \rightarrow Z \rightarrow X \rightarrow C \rightarrow A$ and is invalid. Adding an intervention on X breaks the cycle but also removes the shared latent information; it is not a reward-preserving repair of the reduced policy. The original proposition therefore cannot be retained with only its proof changed. In the corrected proposition $\mathbf{D} = \{Z\}$, but $\mathcal{F}$ continues to exclude the invalid scope. This constraint is essential to the asserted equality of NRO and POMPS sets.

If desired, the validity constraint can itself be stored on $\mathbf{R}$: retain the original directed edges on $\mathbf{R}$ and add a directed edge for every path whose internal vertices lie in $\mathbf{D}$. Removing incoming action edges and adding scope edges in this auxiliary directed graph tests exactly the original acyclicity, because deleted vertices are neither actions nor retained contexts. This graph is used only for feasibility; the reduced SCM uses $\mathcal{H} = \mathcal{G}[\mathbf{R}]$.

Changes

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Original publication. Sanghack Lee and Elias Bareinboim. *Characterizing Optimal Mixed Policies: Where to Intervene and What to Observe*. Advances in Neural Information Processing Systems 33, 2020. Proceedings record. Corrections use the November 2020 R-63 reprint as the baseline. Prepared with AI assistance. These proposed corrections are unofficial and are not a publisher-issued correction or author approval.

Location	Before → After	Assumptions / limits
Preliminaries; Thm. 1	Implicit domain conventions → state finite endogenous domains and the model class for the graphical characterization.	Thm. 1 permits sufficiently large finite domains and unused structural inputs; sufficiency for fixed cardinalities is not asserted. Exogenous domains may be general.
Alg. 2; Thm. 4	Unique maximal NR scope → unique minimal graphical reduction, independent of processing order, with prescribed reward preservation.	Original one-pass algorithm retained. Uniqueness is among graphical reductions. NR uses Thm. 1’s model class.
§4; Def. 5; Prop. 4	Identify POMPS with maximality; use weak improvement to exclude candidates → distinguish the two properties and require a distinct NRO comparator for exclusion.	The maximality-to-POMPS converse is not established. General coverage of an optimal scope by the POMPS set is not established.
Prop. 5; App. F	Preserve POMPS by unrestricted graph and variable pruning → delete unavoidable reward descendants while preserving the original scope-validity constraint.	All eligible actions are retained. NRO and POMPS are preserved in the constrained reduced problem, with retained domains unchanged.
App. C	Extend all results directly to action-specific observations → require availability of any observations introduced by policy substitution.	The original common set of available contexts remains the setting of Prop. 2.
Throughout	Expand supporting proofs and correct local formulas, notation, terminology, and references; clarify the experimental discussion in App. A.	Independent policy randomness and shared latent dependence are retained. No added positivity or ordering assumption.

Experimental scope. Original simulations were not rerun and plot values are unchanged. Exact calculations confirm the two printed SCMs’ optimal means: introduction 1 versus CB 3/4; Appendix A 41/50 versus CB 1/2. The original simulator was unavailable for verification.

Citation. Cite the original publication. If using this version, add a bibliography *note* identifying the *Unofficial AI-corrected version dated 2026-09-20* consulted and this URL: https://www.sanghacklee.me/ai-corrected/POMPS2020_AI_corrected_2026-09-20.pdf. Retain the original publication year.

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